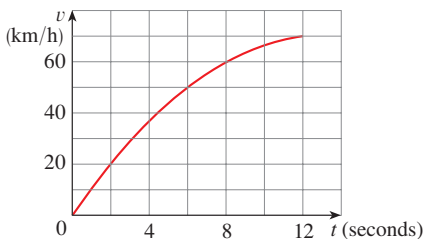


12. The velocity graph of an accelerating car is shown.
- Use the Midpoint rule to estimate the average velocity of the car during the first 12 seconds.
  - At what time was the instantaneous velocity equal to the average velocity?



13. In a certain city the temperature (in  $^{\circ}\text{F}$ )  $t$  hours after 9 AM was modeled by the function

$$T(t) = 50 + 14 \sin \frac{\pi t}{12}$$

Find the average temperature during the period from 9 AM to 9 PM.

14. If a cup of coffee has temperature  $95^{\circ}\text{C}$  in a room where the temperature is  $20^{\circ}\text{C}$ , then, according to Newton's Law of Cooling, the temperature of the coffee after  $t$  minutes is  $T(t) = 20 + 75e^{-t/50}$ . What is the average temperature of the coffee during the first half hour?
15. The **population of Indonesia** from 1950 to 2000 has been modeled with the function

$$P(t) = 83e^{0.18t}$$

where  $P$  is measured in millions and  $t$  is measured in years with  $t = 0$  in the year 1950. What was the average population of Indonesia in the second half of the 20th century?

16. **Blood alcohol concentration** In Section 3.1 we modeled the BAC of male adult subjects after rapid consumption of 15 mL of ethanol (corresponding to one alcoholic drink) by the concentration function

$$C(t) = 0.0225te^{-0.0467t}$$

where  $t$  is measured in minutes after consumption and  $C(t)$  is measured in mg/mL. What was the average BAC during the first hour?

17. **Measles pathogenesis** In Section 5.1 we modeled the infection level of the measles virus in a patient by the function

$$f(t) = -t(t - 21)(t + 1)$$

where  $t$  is measured in days and  $f(t)$  is measured in the number of infected cells per mL of blood plasma. Over the course of the 21-day infection, what is the average level of infection?

18. **Length of a fish** For a fish that starts life with a length of 1 cm and has a maximum length of 30 cm, the von Bertalanffy growth model predicts that the growth rate is  $29e^{-a}$  cm/year. What is the average length of the fish over its first five years?
19. **Breathing** is cyclic and a full respiratory cycle from the beginning of inhalation to the end of exhalation takes about 5 s. The maximum rate of air flow into the lungs is about 0.5 L/s. This explains, in part, why the function

$$f(t) = \frac{1}{2} \sin \frac{2\pi t}{5}$$

has often been used to model the rate of air flow into the lungs. If inhalation occurs during the interval  $0 \leq t \leq 2.5$ , what is the average rate of air flow during inhalation?

20. **Blood flow** The velocity  $v$  of blood that flows in a blood vessel with radius  $R$  and length  $l$  at a distance  $r$  from the central axis is

$$v(r) = \frac{P}{4\eta l} (R^2 - r^2)$$

where  $P$  is the pressure difference between the ends of the vessel and  $\eta$  is the viscosity of the blood (see Example 3.3.9). Find the average velocity (with respect to  $r$ ) over the interval  $0 \leq r \leq R$ . Compare the average velocity with the maximum velocity.

21. If  $f$  is continuous and  $\int_1^3 f(x) dx = 8$ , show that  $f$  takes on the value 4 at least once on the interval  $[1, 3]$ .
22. Find the numbers  $b$  such that the average value of  $f(x) = 2 + 6x - 3x^2$  on the interval  $[0, b]$  is equal to 3.
23. Prove the Mean Value Theorem for Integrals by applying the Mean Value Theorem for derivatives (see Section 4.2) to the function  $F(x) = \int_a^x f(t) dt$ .

## 6.3 Further Applications to Biology

In Chapter 5 and in Sections 6.1 and 6.2 we presented several applications of integration to biology. In this section we consider additional examples of how integrals are used in biological settings: survival and renewal of populations, blood flow in veins and arteries, and cardiac output. Other applications are explored in the exercises.

### ■ Survival and Renewal

A population may be continually adding members while some of the existing members die. If we can model how these changes occur with suitable functions, we can predict the population size at any point in the future.

Suppose we start with an initial population  $P_0$  and new members are added at the rate  $R(t)$ , where  $t$  is the number of years from now. We call  $R$  a **renewal function**. In addition, the proportion of the population that survives at least  $t$  years from now is given by a **survival function**  $S$ . [So if  $S(5) = 0.8$ , 80% of the current population remains after 5 years.]

To predict the population in  $T$  years, we first note that  $S(t) \cdot P_0$  members of the current population survive. To account for the newly added members, we divide the time interval  $[0, T]$  into  $n$  subintervals, each of length  $\Delta t = T/n$ , and let  $t_i$  be the right endpoint of the  $i$ th subinterval. During this time interval, approximately  $R(t_i) \Delta t$  members are added, and the proportion of them that survive until time  $T$  is given by  $S(T - t_i)$ . Thus the remaining members of those added during this time interval is

$$(\text{proportion surviving})(\text{number of members}) = S(T - t_i) R(t_i) \Delta t$$

Then the total number of new members to the population who survive after  $T$  years is approximately

$$\sum_{i=1}^n S(T - t_i) R(t_i) \Delta t$$

If we let  $n \rightarrow \infty$ , this Riemann sum approaches the integral

$$\int_0^T S(T - t) R(t) dt$$

By adding this integral to the number of initial members who survived, we get the total population after  $T$  years.

A population begins with  $P_0$  members and members are added at a rate given by the renewal function  $R(t)$ , where  $t$  is measured in years. The proportion of the population that remains after  $t$  years is given by the survival function  $S(t)$ . Then the population  $T$  years from now is given by

$$(1) \quad P(T) = S(T) \cdot P_0 + \int_0^T S(T - t) R(t) dt$$

Equation 1 is also valid if  $t$  represents any other unit of time, such as weeks or months.

**EXAMPLE 1** | Predicting a future population There are currently 5600 trout in a lake and the trout are reproducing at the rate  $R(t) = 720e^{0.1t}$  fish/year. However, pollution is killing many of the trout; the proportion that survive after  $t$  years is given by  $S(t) = e^{-0.2t}$ . How many trout will there be in the lake in 10 years?

**SOLUTION** We have  $P_0 = 5600$  and  $T = 10$ , so by Formula 1, the population in ten years is

$$\begin{aligned} P(10) &= S(10) \cdot 5600 + \int_0^{10} S(10 - t) R(t) dt \\ &= 5600e^{-0.2(10)} + \int_0^{10} e^{-0.2(10-t)} \cdot 720e^{0.1t} dt \\ &= 5600e^{-2} + 720 \int_0^{10} e^{0.3t-2} dt \end{aligned}$$

Writing  $e^{0.3t-2}$  as  $e^{0.3t}e^{-2}$  gives

$$\begin{aligned}
 P(10) &= 5600e^{-2} + 720e^{-2} \int_0^{10} e^{0.3t} dt \\
 &= 5600e^{-2} + 720e^{-2} \left. \frac{e^{0.3t}}{0.3} \right|_0^{10} \\
 &= 5600e^{-2} + \frac{720}{0.3}e^{-2}(e^3 - e^0) \\
 &= 5600e^{-2} + 2400(e - e^{-2}) \approx 6956.95
 \end{aligned}$$

Thus we predict that there will be about 6960 trout in the lake ten years from now. ■

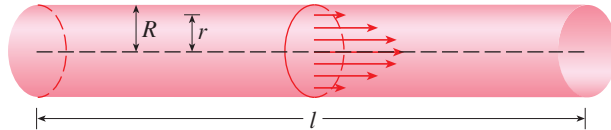
Although we presented Formula 1 in the context of populations, it applies to other settings as well, such as administering a drug over time as the body works to eliminate the drug. The exercises investigate additional applications.

### ■ Blood Flow

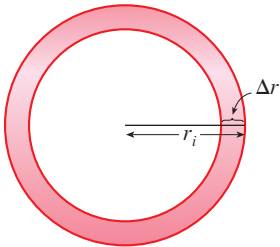
In Example 3.3.9 we discussed the law of laminar flow:

$$v(r) = \frac{P}{4\eta l} (R^2 - r^2)$$

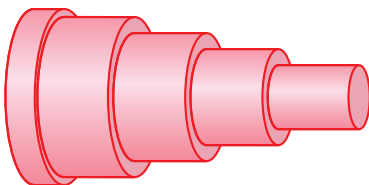
which gives the velocity  $v$  of blood that flows along a blood vessel with radius  $R$  and length  $l$  at a distance  $r$  from the central axis, where  $P$  is the pressure difference between the ends of the vessel and  $\eta$  is the viscosity of the blood (see Figure 1).



**FIGURE 1**  
Blood flow in an artery



**FIGURE 2**



**FIGURE 3**

In order to compute the rate of blood flow, or *flux* (volume per unit time), we consider smaller, equally spaced radii  $r_1, r_2, \dots$ . The approximate area of the ring with inner radius  $r_{i-1}$  and outer radius  $r_i$  is

$$2\pi r_i \Delta r \quad \text{where} \quad \Delta r = r_i - r_{i-1}$$

(See Figure 2.) If  $\Delta r$  is small, then the velocity is almost constant throughout this ring and can be approximated by  $v(r_i)$ . Thus the volume of blood per unit time that flows across the ring is approximately

$$(2\pi r_i \Delta r) v(r_i) = 2\pi r_i v(r_i) \Delta r$$

and the total volume of blood that flows across a cross-section per unit time is about

$$\sum_{i=1}^n 2\pi r_i v(r_i) \Delta r$$

This approximation is illustrated in Figure 3. Notice that the velocity (and hence the volume per unit time) increases toward the center of the blood vessel. The approximation gets better as  $n$  increases. When we take the limit as  $n \rightarrow \infty$  we get an integral that gives the exact value of the **flux** (or *discharge*), which is the volume of blood that passes

a cross-section per unit time:

$$\begin{aligned}
 F &= \int_0^R 2\pi r v(r) dr \\
 &= \int_0^R 2\pi r \frac{P}{4\eta l} (R^2 - r^2) dr \\
 &= \frac{\pi P}{2\eta l} \int_0^R (R^2 r - r^3) dr = \frac{\pi P}{2\eta l} \left[ R^2 \frac{r^2}{2} - \frac{r^4}{4} \right]_{r=0}^{r=R} \\
 &= \frac{\pi P}{2\eta l} \left[ \frac{R^4}{2} - \frac{R^4}{4} \right] = \frac{\pi P R^4}{8\eta l}
 \end{aligned}$$

The resulting equation

$$(2) \quad F = \frac{\pi P R^4}{8\eta l}$$

is called **Poiseuille's Law**; it shows that the flux is proportional to the fourth power of the radius of the blood vessel.

In Exercise 10 you are asked to investigate the effect on blood pressure if the radius of an artery is reduced to three-fourths of its normal value.

### ■ Cardiac Output

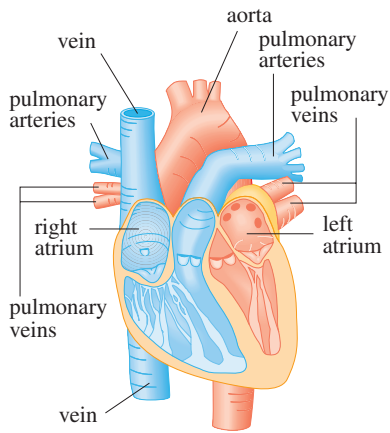


FIGURE 4

Figure 4 shows the human heart and associated blood vessels. Blood returns from the body through the veins, enters the right atrium of the heart, and is pumped to the lungs through the pulmonary arteries for oxygenation. It then flows back into the left atrium through the pulmonary veins and then out to the rest of the body through the aorta. The **cardiac output** of the heart is the volume of blood pumped by the heart per unit time, that is, the rate of flow into the aorta.

The *dye dilution method* is used to measure the cardiac output. Dye is injected into the right atrium and flows through the heart into the aorta. A probe inserted into the aorta measures the concentration of the dye leaving the heart at equally spaced times over a time interval  $[0, T]$  until the dye has cleared. Let  $c(t)$  be the concentration of the dye at time  $t$ . If we divide  $[0, T]$  into subintervals of equal length  $\Delta t$ , then the amount of dye that flows past the measuring point during the subinterval from  $t = t_{i-1}$  to  $t = t_i$  is approximately

$$(\text{concentration of dye in blood})(\text{volume of blood}) = c(t_i)(F \Delta t)$$

where  $F$  is the rate of flow that we are trying to determine. Thus the total amount of dye is approximately

$$c(t_1)F\Delta t + c(t_2)F\Delta t + \cdots + c(t_n)F\Delta t$$

and, letting  $n \rightarrow \infty$ , we find that the amount of dye is

$$A = \int_0^T c(t)F dt = F \int_0^T c(t) dt$$

Thus the cardiac output is given by

$$(3) \quad F = \frac{A}{\int_0^T c(t) dt}$$

where the amount of dye  $A$  is known and the integral can be approximated from the concentration readings.

| $t$ | $c(t)$ | $t$ | $c(t)$ |
|-----|--------|-----|--------|
| 0   | 0      | 6   | 6.1    |
| 1   | 0.4    | 7   | 4.0    |
| 2   | 2.8    | 8   | 2.3    |
| 3   | 6.5    | 9   | 1.1    |
| 4   | 9.8    | 10  | 0      |
| 5   | 8.9    |     |        |

**EXAMPLE 2 | Cardiac output** A 5-mg dose (called a bolus) of dye is injected into a right atrium. The concentration of the dye (in milligrams per liter) is measured in the aorta at one-second intervals as shown in the chart. Estimate the cardiac output.

**SOLUTION** Here  $A = 5$  and  $T = 10$ . We can use the Midpoint Rule with  $n = 5$  subdivisions to approximate the integral of the concentration. Then  $\Delta t = 2$  and

$$\begin{aligned}\int_0^{10} c(t) dt &\approx [c(1) + c(3) + c(5) + c(7) + c(9)] \Delta t \\ &= [0.4 + 6.5 + 8.9 + 4.0 + 1.1] (2) \\ &= 41.8\end{aligned}$$

Thus Formula 3 gives the cardiac output to be

$$F = \frac{A}{\int_0^{10} c(t) dt} \approx \frac{5}{41.8} \approx 0.12 \text{ L/s} = 7.2 \text{ L/min}$$

## EXERCISES 6.3

- Animal survival and renewal** An animal population currently has 7400 members and is reproducing at the rate  $R(t) = 2240 + 60t$  members/year. The proportion of members that survive after  $t$  years is given by  $S(t) = 1/(t + 1)$ .
  - How many of the original members survive four years?
  - How many new members are added during the next four years?
  - Explain why the animal population four years from now is not the same as the sum of your answers from parts (a) and (b).
- City population** A city currently has 36,000 residents and is adding new residents steadily at the rate of 1600 per year. If the proportion of residents that remain after  $t$  years is given by  $S(t) = 1/(t + 1)$ , what is the population of the city seven years from now?
- Insect survival and renewal** A population of insects currently numbers 22,500 and is increasing at a rate of  $R(t) = 1225e^{0.14t}$  insects/week. If the survival function for the insects is  $S(t) = e^{-0.2t}$ , where  $t$  is measured in weeks, how many insects are there after 12 weeks?
- Animal survival and renewal** There are currently 3800 birds of a particular species in a national park and their number is increasing at a rate of  $R(t) = 525e^{0.05t}$  birds/year. If the proportion of birds that survive  $t$  years is given by  $S(t) = e^{-0.1t}$ , what do you predict the bird population will be 10 years from now?
- Drug concentration** A drug is administered intravenously to a patient at the rate of 12 mg/h. The patient's body eliminates the drug over time so that after  $t$  hours the proportion that remains is  $e^{-0.25t}$ . If the patient currently has 50 mg of the drug in her bloodstream, how much of the drug is present eight hours from now?
- Drug concentration** A patient receives a drug at a constant rate of 30 mg/h. The drug is eliminated from the

bloodstream over time so that the fraction  $e^{-0.2t}$  remains after  $t$  hours. The patient currently has 80 mg of the drug present in the bloodstream. How much will be present in 24 hours?

- Water pollution** A contaminant is leaking into a lake at a rate of

$$R(t) = 1600e^{0.06t} \text{ gallons/h}$$

Enzymes have been added to the lake that neutralize the contaminant over time so that after  $t$  hours the fraction of the contaminant that remains is  $S(t) = e^{-0.32t}$ . If there are currently 10,000 gallons of the contaminant in the lake, how many gallons are present in the lake 18 hours from now?

- Insect survival and renewal** Sterile fruit flies are used in an experiment where the proportion that survive at least  $t$  days is given by  $e^{-0.15t}$ . If the experiment begins with 200 fruit flies, and flies are added at the rate of 5 per hour, how many flies are present 14 days after the start of the experiment?
- Blood flow** Use Poiseuille's Law to calculate the rate of flow in a small human artery where we can take  $\eta = 0.027 \text{ dyn} \cdot \text{s/cm}^2$ ,  $R = 0.008 \text{ cm}$ ,  $l = 2 \text{ cm}$ , and  $P = 4000 \text{ dyn/cm}^2$ .
- Blood flow** High blood pressure results from constriction of the arteries. To maintain a normal flow rate (flux), the heart has to pump harder, thus increasing the blood pressure. Use Poiseuille's Law to show that if  $R_0$  and  $P_0$  are normal values of the radius and pressure in an artery and the constricted values are  $R$  and  $P$ , then for the flux to remain constant,  $P$  and  $R$  are related by the equation

$$\frac{P}{P_0} = \left(\frac{R_0}{R}\right)^4$$

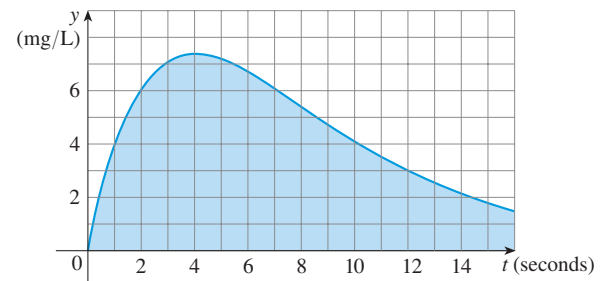
Deduce that if the radius of an artery is reduced to three-

fourths of its former value, then the pressure is more than tripled.

- 11. Cardiac output** The dye dilution method is used to measure cardiac output with 6 mg of dye. The dye concentrations, in mg/L, are modeled by  $c(t) = 20te^{-0.6t}$ ,  $0 \leq t \leq 10$ , where  $t$  is measured in seconds. Find the cardiac output. [Hint: Integration by parts is required.]
- 12. Cardiac output** After an 8-mg injection of dye, the readings of dye concentration, in mg/L, at two-second intervals are as shown in the table. Use the Midpoint Rule to estimate the cardiac output.

| $t$ | $c(t)$ | $t$ | $c(t)$ |
|-----|--------|-----|--------|
| 0   | 0      | 12  | 3.9    |
| 2   | 2.4    | 14  | 2.3    |
| 4   | 5.1    | 16  | 1.6    |
| 6   | 7.8    | 18  | 0.7    |
| 8   | 7.6    | 20  | 0      |
| 10  | 5.4    |     |        |

- 13. Cardiac output** The graph of the concentration function  $c(t)$  is shown after a 7-mg injection of dye into a heart. Use the Midpoint Rule to estimate the cardiac output.



- 14. Drug administration** A patient is continually receiving a drug. If the drug is eliminated from the body over time so that the fraction that remains after  $t$  hours is  $e^{-0.4t}$ , at what constant rate should the drug be administered to maintain a steady level of the drug in the bloodstream?

## 6.4 Volumes

In trying to find the volume of a solid we face the same type of problem as in finding areas. We have an intuitive idea of what volume means, but we must make this idea precise by using calculus to give an exact definition of volume.

We start with a simple type of solid called a **cylinder** (or, more precisely, a *right cylinder*). As illustrated in Figure 1(a), a cylinder is bounded by a plane region  $B_1$ , called the base, and a congruent region  $B_2$  in a parallel plane. The cylinder consists of all points on line segments that are perpendicular to the base and join  $B_1$  to  $B_2$ . If the area of the base is  $A$  and the height of the cylinder (the distance from  $B_1$  to  $B_2$ ) is  $h$ , then the volume  $V$  of the cylinder is defined as

$$V = Ah$$

In particular, if the base is a circle with radius  $r$ , then the cylinder is a circular cylinder with volume  $V = \pi r^2 h$  [see Figure 1(b)], and if the base is a rectangle with length  $l$  and width  $w$ , then the cylinder is a rectangular box (also called a *rectangular parallelepiped*) with volume  $V = lwh$  [see Figure 1(c)].

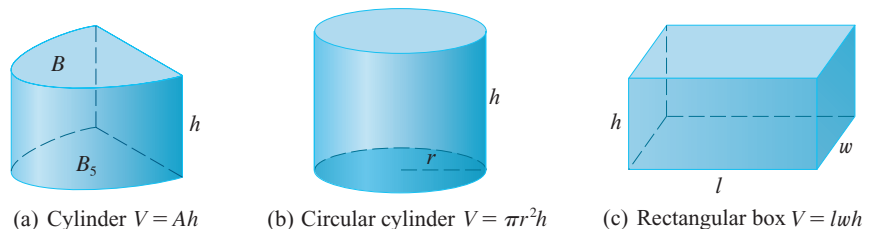
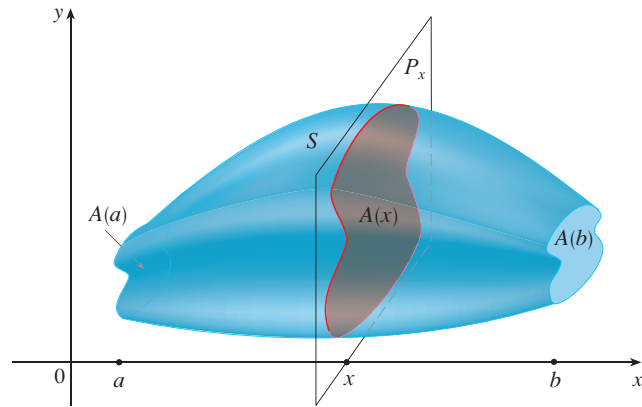


FIGURE 1

For a solid  $S$  that isn't a cylinder we first "cut"  $S$  into pieces and approximate each piece by a cylinder. We estimate the volume of  $S$  by adding the volumes of the cylinders. We arrive at the exact volume of  $S$  through a limiting process in which the number of pieces becomes large.

We start by intersecting  $S$  with a plane and obtaining a plane region that is called a **cross-section** of  $S$ . Let  $A(x)$  be the area of the cross-section of  $S$  in a plane  $P_x$  perpendicular to the  $x$ -axis and passing through the point  $x$ , where  $a \leq x \leq b$ . (See Figure 2. Think of slicing  $S$  with a knife through  $x$  and computing the area of this slice.) The cross-sectional area  $A(x)$  will vary as  $x$  increases from  $a$  to  $b$ .

FIGURE 2



Let's divide  $S$  into  $n$  "slabs" of equal width  $\Delta x$  by using the planes  $P_{x_1}, P_{x_2}, \dots$  to slice the solid. (Think of slicing a loaf of bread.) If we choose sample points  $x_i^*$  in  $[x_{i-1}, x_i]$ , we can approximate the  $i$ th slab  $S_i$  (the part of  $S$  that lies between the planes  $P_{x_{i-1}}$  and  $P_{x_i}$ ) by a cylinder with base area  $A(x_i^*)$  and "height"  $\Delta x$ . (See Figure 3.)

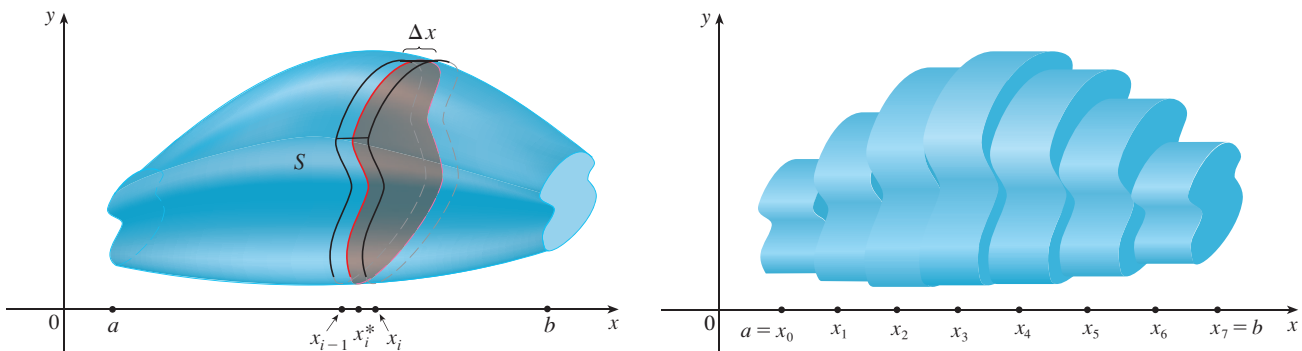


FIGURE 3

The volume of this cylinder is  $A(x_i^*) \Delta x$ , so an approximation to our intuitive conception of the volume of the  $i$ th slab  $S_i$  is

$$V(S_i) \approx A(x_i^*) \Delta x$$

Adding the volumes of these slabs, we get an approximation to the total volume (that is, what we think of intuitively as the volume):

$$V \approx \sum_{i=1}^n A(x_i^*) \Delta x$$

This approximation appears to become better and better as  $n \rightarrow \infty$ . (Think of the slices as becoming thinner and thinner.) Therefore we *define* the volume as the limit of these

sums as  $n \rightarrow \infty$ . But we recognize the limit of Riemann sums as a definite integral and so we have the following definition.

It can be proved that this definition is independent of how  $S$  is situated with respect to the  $x$ -axis. In other words, no matter how we slice  $S$  with parallel planes, we always get the same answer for  $V$ .

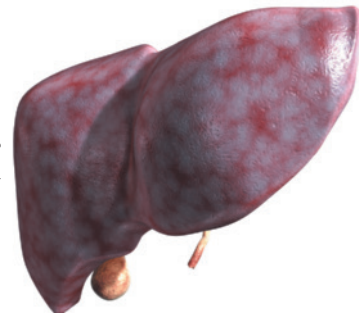
**Definition of Volume** Let  $S$  be a solid that lies between  $x = a$  and  $x = b$ . If the cross-sectional area of  $S$  in the plane  $P_x$ , through  $x$  and perpendicular to the  $x$ -axis, is  $A(x)$ , where  $A$  is a continuous function, then the **volume** of  $S$  is

$$V = \lim_{n \rightarrow \infty} \sum_{i=1}^n A(x_i^*) \Delta x = \int_a^b A(x) dx$$

When we use the volume formula  $V = \int_a^b A(x) dx$ , it is important to remember that  $A(x)$  is the area of a moving cross-section obtained by slicing through  $x$  perpendicular to the  $x$ -axis.

Notice that, for a cylinder, the cross-sectional area is constant:  $A(x) = A$  for all  $x$ . So our definition of volume gives  $V = \int_a^b A dx = A(b - a)$ ; this agrees with the formula  $V = Ah$ .

MedicalRF.com / Getty Images

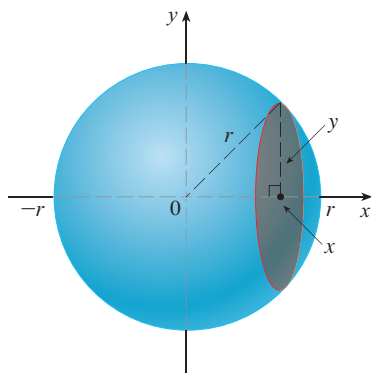


**EXAMPLE 1** | **Volume of a liver** A CAT scan produces equally spaced cross-sectional views of a human organ that provide information about the organ otherwise obtained only by surgery. For example, such measurements of liver volume can be related to diseases such as cirrhosis.<sup>1</sup> Suppose that a CAT scan of a human liver shows cross-sections spaced 2 cm apart. The liver is 20 cm long and the cross-sectional areas, in square centimeters, are 0, 39, 63, 128, 117, 106, 94, 79, 58, 18, and 0. Use the Midpoint Rule to estimate the volume of the liver.

**SOLUTION** In using the Midpoint Rule we will use  $n = 5$  subintervals; their midpoints are 2, 6, 10, 14, and 18. If  $A(x)$  is the area of the cross-section of the liver at a distance of  $x$  centimeters from one end, then the volume is

$$\begin{aligned} V &= \int_0^{20} A(x) dx \\ &= \frac{20 - 0}{5} [A(2) + A(6) + A(10) + A(14) + A(18)] \\ &= 4[39 + 128 + 106 + 79 + 18] \\ &= 4 \cdot 370 = 1480 \end{aligned}$$

The volume of the liver is approximately  $1480 \text{ cm}^3$ . ■



**FIGURE 4**

**EXAMPLE 2** | Show that the volume of a sphere of radius  $r$  is  $V = \frac{4}{3}\pi r^3$ .

**SOLUTION** If we place the sphere so that its center is at the origin (see Figure 4), then the plane  $P_x$  intersects the sphere in a circle whose radius (from the Pythagorean Theorem) is  $y = \sqrt{r^2 - x^2}$ . So the cross-sectional area is

$$A(x) = \pi y^2 = \pi(r^2 - x^2)$$

1. J.-Y. Zhu et al., "Measurement of Liver Volume and Its Clinical Significance in Cirrhotic Portal Hypertensive Patients," *World Journal of Gastroenterology* 5 (1999): 525–26.

Using the definition of volume with  $a = -r$  and  $b = r$ , we have

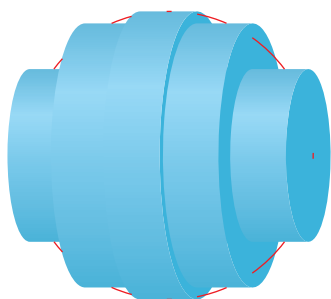
$$\begin{aligned}
 V &= \int_{-r}^r A(x) \, dx = \int_{-r}^r \pi(r^2 - x^2) \, dx \\
 &= 2\pi \int_0^r (r^2 - x^2) \, dx && \text{(The integrand is even.)} \\
 &= 2\pi \left[ r^2x - \frac{x^3}{3} \right]_0^r = 2\pi \left( r^3 - \frac{r^3}{3} \right) \\
 &= \frac{4}{3}\pi r^3
 \end{aligned}$$

Figure 5 illustrates the definition of volume when the solid is a sphere with radius  $r = 1$ . From the result of Example 2, we know that the volume of the sphere is  $\frac{4}{3}\pi$ , which is approximately 4.18879. Here the slabs are circular cylinders, or *disks*, and the three parts of Figure 5 show the geometric interpretations of the Riemann sums

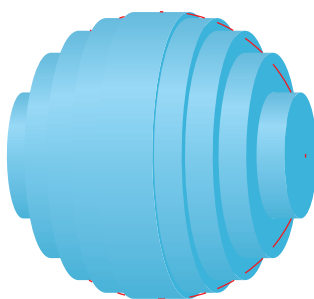
$$\sum_{i=1}^n A(\bar{x}_i) \Delta x = \sum_{i=1}^n \pi(1^2 - \bar{x}_i^2) \Delta x$$

**TEC** Visual 6.4A shows an animation of Figure 5.

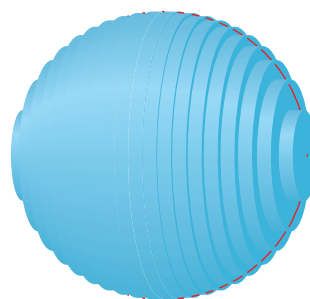
when  $n = 5, 10$ , and  $20$  if we choose the sample points  $x_i^*$  to be the midpoints  $\bar{x}_i$ . Notice that as we increase the number of approximating cylinders, the corresponding Riemann sums become closer to the true volume.



(a) Using 5 disks,  $V \approx 4.2726$

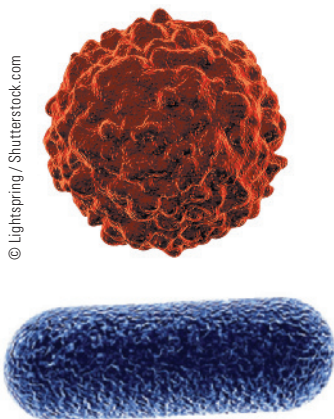


(b) Using 10 disks,  $V \approx 4.2097$



(c) Using 20 disks,  $V \approx 4.1940$

**FIGURE 5** Approximating the volume of a sphere with radius 1



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You have probably seen the formula  $V = \frac{4}{3}\pi r^3$  for the volume of a sphere before. But calculus is required to prove it, so this may be the first time you have seen it proved.

This formula is useful in biology because the shape of a tumor is often modeled by a sphere. (See, for instance, Example 3.5.14.) And the shape of a bacterium can often be modeled as a circular cylinder capped by two hemispheres as shown in the margin.

**EXAMPLE 3** | Find the volume of the solid obtained by rotating about the  $x$ -axis the region under the curve  $y = \sqrt{x}$  from 0 to 1. Illustrate the definition of volume by sketching a typical approximating cylinder.

**SOLUTION** The region is shown in Figure 6(a). If we rotate about the  $x$ -axis, we get the solid shown in Figure 6(b). When we slice through the point  $x$ , we get a disk with radius  $\sqrt{x}$ . The area of this cross-section is

$$A(x) = \pi(\sqrt{x})^2 = \pi x$$

and the volume of the approximating cylinder (a disk with thickness  $\Delta x$ ) is

$$A(x) \Delta x = \pi x \Delta x$$

The solid lies between  $x = 0$  and  $x = 1$ , so its volume is

$$V = \int_0^1 A(x) dx = \int_0^1 \pi x dx = \pi \left[ \frac{x^2}{2} \right]_0^1 = \frac{\pi}{2}$$

Did we get a reasonable answer in Example 3? As a check on our work, let's replace the given region by a square with base  $[0, 1]$  and height 1. If we rotate this square, we get a cylinder with radius 1, height 1, and volume  $\pi \cdot 1^2 \cdot 1 = \pi$ . We computed that the given solid has half this volume. That seems about right.

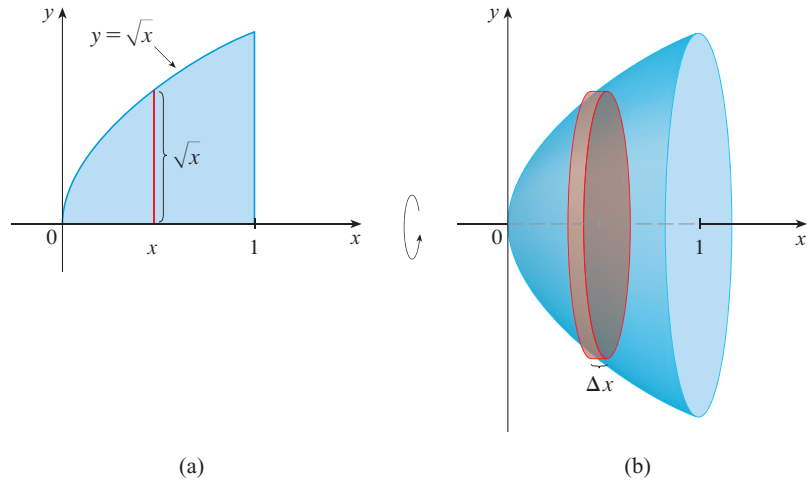


FIGURE 6

**EXAMPLE 4** | The region enclosed by the curves  $y = x$  and  $y = x^2$  is rotated about the  $x$ -axis. Find the volume of the resulting solid.

**SOLUTION** The curves  $y = x$  and  $y = x^2$  intersect at the points  $(0, 0)$  and  $(1, 1)$ . The region between them, the solid of rotation, and a cross-section perpendicular to the  $x$ -axis are shown in Figure 7. A cross-section in the plane  $P_x$  has the shape of a *washer* (an annular ring) with inner radius  $x^2$  and outer radius  $x$ , so we find the cross-sectional area by subtracting the area of the inner circle from the area of the outer circle:

$$A(x) = \pi x^2 - \pi (x^2)^2 = \pi (x^2 - x^4)$$

**TEC** Visual 6.4B shows how solids of revolution are formed.

Therefore we have

$$V = \int_0^1 A(x) dx = \int_0^1 \pi (x^2 - x^4) dx = \pi \left[ \frac{x^3}{3} - \frac{x^5}{5} \right]_0^1 = \frac{2\pi}{15}$$

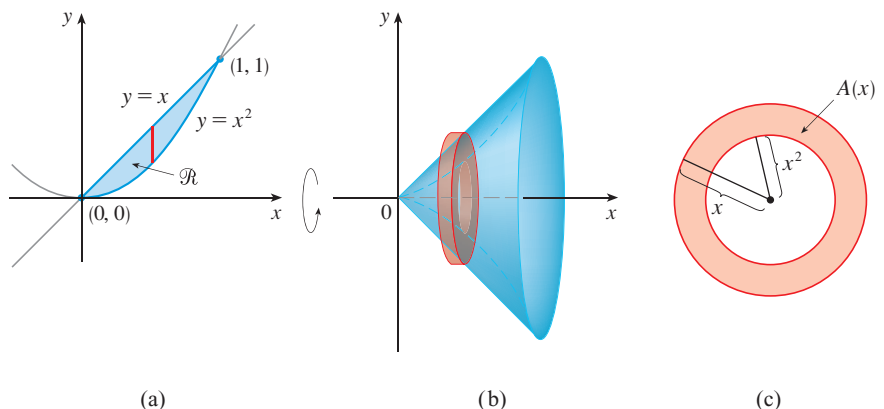


FIGURE 7

The solids in Examples 2–4 are all called **solids of revolution** because they are obtained by revolving a region about a line. In general, we calculate the volume of a solid of revolution by using the basic defining formula

$$V = \int_a^b A(x) \, dx$$

and we find the cross-sectional area  $A(x)$  in one of the following ways:

- If the cross-section is a disk (as in Examples 2 and 3), we find the radius of the disk (in terms of  $x$ ) and use

$$A = \pi(\text{radius})^2$$

- If the cross-section is a washer (as in Example 4), we find the inner radius  $r_{\text{in}}$  and outer radius  $r_{\text{out}}$  from a sketch (as in Figures 7 and 8) and compute the area of the washer by subtracting the area of the inner disk from the area of the outer disk:

$$A = \pi(\text{outer radius})^2 - \pi(\text{inner radius})^2$$

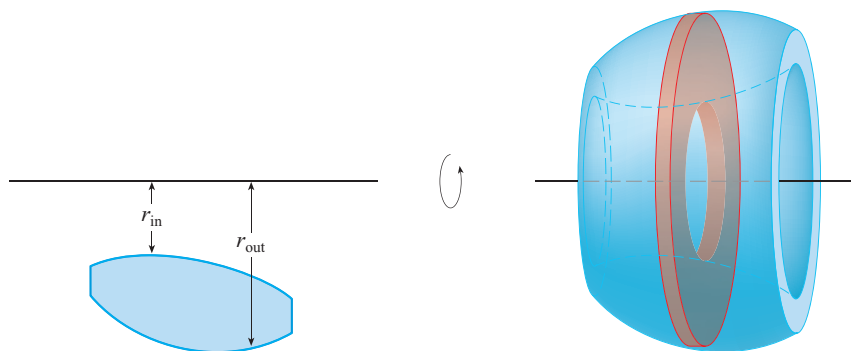


FIGURE 8

We now find the volume of a solid that is *not* a solid of revolution.

**TEC** Visual 6.4C shows how the solid in Figure 9 is generated.

**EXAMPLE 5** | Figure 9 shows a solid with a circular base of radius 1. Parallel cross-sections perpendicular to the base are equilateral triangles. Find the volume of the solid.

**SOLUTION** Let's take the circle to be  $x^2 + y^2 = 1$ . The solid, its base, and a typical cross-section at a distance  $x$  from the origin are shown in Figure 10.

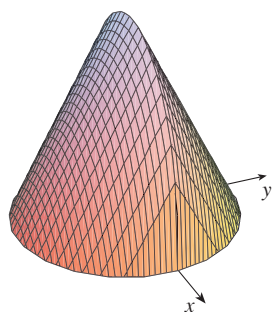


FIGURE 9

Computer-generated picture of the solid in Example 5

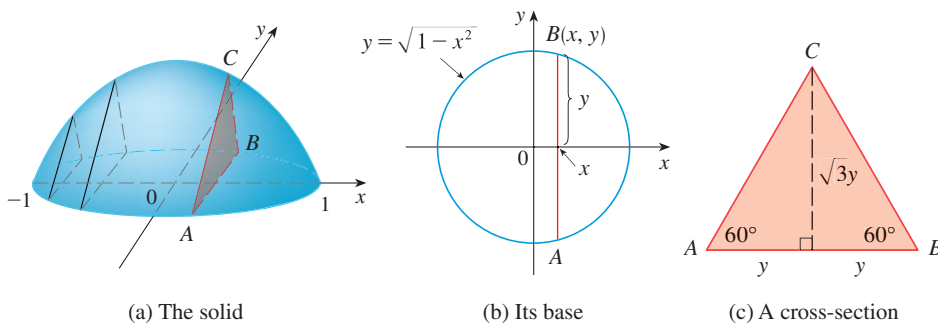


FIGURE 10

Since  $B$  lies on the circle, we have  $y = \sqrt{1 - x^2}$  and so the base of the triangle  $ABC$  is  $|AB| = 2\sqrt{1 - x^2}$ . Since the triangle is equilateral, we see from Figure 10(c)

that its height is  $\sqrt{3}y = \sqrt{3}\sqrt{1-x^2}$ . The cross-sectional area is therefore

$$A(x) = \frac{1}{2} \cdot 2\sqrt{1-x^2} \cdot \sqrt{3}\sqrt{1-x^2} = \sqrt{3}(1-x^2)$$

and the volume of the solid is

$$\begin{aligned} V &= \int_{-1}^1 A(x) dx = \int_{-1}^1 \sqrt{3}(1-x^2) dx \\ &= 2 \int_0^1 \sqrt{3}(1-x^2) dx = 2\sqrt{3} \left[ x - \frac{x^3}{3} \right]_0^1 = \frac{4\sqrt{3}}{3} \end{aligned}$$

## EXERCISES 6.4

**1–6** Find the volume of the solid obtained by rotating the region bounded by the given curves about the  $x$ -axis. Sketch the region, the solid, and a typical disk or washer.

1.  $y = 2 - \frac{1}{2}x$ ,  $y = 0$ ,  $x = 1$ ,  $x = 2$

2.  $y = 1 - x^2$ ,  $y = 0$

3.  $y = \sqrt{x-1}$ ,  $y = 0$ ,  $x = 5$

4.  $y = \sqrt{25-x^2}$ ,  $y = 0$ ,  $x = 2$ ,  $x = 4$

5.  $y = x^3$ ,  $y = x$ ,  $x \geq 0$

6.  $y = \frac{1}{4}x^2$ ,  $y = 5 - x^2$

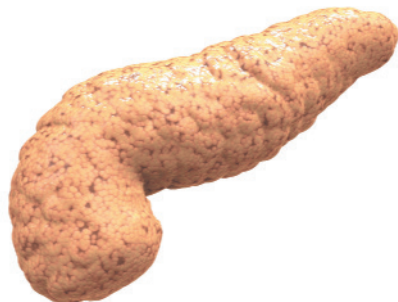
**7–8** Here we rotate about the  $y$ -axis instead of the  $x$ -axis. Find the volume of the solid obtained by rotating the region bounded by the given curves about the  $y$ -axis. Sketch the region, the solid, and a typical disk.

7.  $x = 2\sqrt{y}$ ,  $x = 0$ ,  $y = 9$

8.  $y = \ln x$ ,  $y = 1$ ,  $y = 2$ ,  $x = 0$

**9. Volume of a pancreas** A CAT scan of a human pancreas shows cross-sections spaced 1 cm apart. The pancreas is 12 cm long and the cross-sectional areas, in square centimeters, are 0, 7.7, 15.2, 18.0, 10.3, 10.8, 9.7, 8.7, 7.7, 5.5, 4.0, 2.7, and 0. Use the Midpoint Rule to estimate the volume of the pancreas.

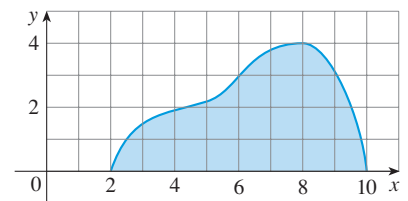
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**10.** A log 10 m long is cut at 1-meter intervals and its cross-sectional areas  $A$  (at a distance  $x$  from the end of the log) are listed in the table. Use the Midpoint Rule with  $n = 5$  to estimate the volume of the log.

| $x$ (m) | $A$ (m <sup>2</sup> ) | $x$ (m) | $A$ (m <sup>2</sup> ) |
|---------|-----------------------|---------|-----------------------|
| 0       | 0.68                  | 6       | 0.53                  |
| 1       | 0.65                  | 7       | 0.55                  |
| 2       | 0.64                  | 8       | 0.52                  |
| 3       | 0.61                  | 9       | 0.50                  |
| 4       | 0.58                  | 10      | 0.48                  |
| 5       | 0.59                  |         |                       |

**11.** (a) If the region shown in the figure is rotated about the  $x$ -axis to form a solid, use the Midpoint Rule with  $n = 4$  to estimate the volume of the solid.



(b) Estimate the volume if the region is rotated about the  $y$ -axis. Again use the Midpoint Rule with  $n = 4$ .

### 12. Volume of a bird's egg

CAS

(a) A model for the shape of a bird's egg is obtained by rotating about the  $x$ -axis the region under the graph of

$$f(x) = (ax^3 + bx^2 + cx + d)\sqrt{1-x^2}$$

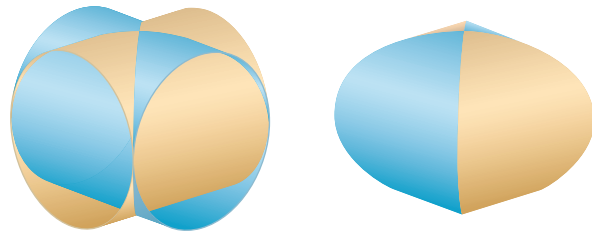
Use a CAS to find the volume of such an egg.

(b) For a red-throated loon,  $a = -0.06$ ,  $b = 0.04$ ,  $c = 0.1$ , and  $d = 0.54$ . Graph  $f$  and find the volume of an egg of this species.

**13–15** Find the volume of the described solid  $S$ .

- 13.**  $S$  is a right circular cone with height  $h$  and base radius  $r$ .
  - 14.** The base of  $S$  is a circular disk with radius  $r$ . Parallel cross-sections perpendicular to the base are squares.
  - 15.** The base of  $S$  is an elliptical region with boundary curve  $9x^2 + 4y^2 = 36$ . Cross-sections perpendicular to the  $x$ -axis are isosceles right triangles with hypotenuse in the base.
- 
- 16.** The base of  $S$  is a circular disk with radius  $r$ . Parallel cross-sections perpendicular to the base are isosceles triangles with height  $h$  and unequal side in the base.
    - (a) Set up an integral for the volume of  $S$ .
    - (b) By interpreting the integral as an area, find the volume of  $S$ .

- 17.** Find the volume common to two spheres, each with radius  $r$ , if the center of each sphere lies on the surface of the other sphere.
- 18.** Find the volume common to two circular cylinders, each with radius  $r$ , if the axes of the cylinders intersect at right angles.



## Chapter 6 REVIEW

### CONCEPT CHECK

- 1.** Draw two typical curves  $y = f(x)$  and  $y = g(x)$ , where  $f(x) \geq g(x)$  for  $a \leq x \leq b$ . Show how to approximate the area between these curves by a Riemann sum and sketch the corresponding approximating rectangles. Then write an expression for the exact area.
- 2.** Suppose that Sue runs faster than Kathy throughout a 1500-meter race. What is the physical meaning of the area between their velocity curves for the first minute of the race?
- 3.** (a) What is the average value of a function  $f$  on an interval  $[a, b]$ ?  
(b) What does the Mean Value Theorem for Integrals say? What is its geometric interpretation?
- 4.** If we have survival and renewal functions for a population, how do we predict the size of the population  $T$  years from now?
- 5.** (a) What is the cardiac output of the heart?  
(b) Explain how the cardiac output can be measured by the dye dilution method.
- 6.** (a) Suppose  $S$  is a solid with known cross-sectional areas. Explain how to approximate the volume of  $S$  by a Riemann sum. Then write an expression for the exact volume.  
(b) If  $S$  is a solid of revolution, how do you find the cross-sectional areas?

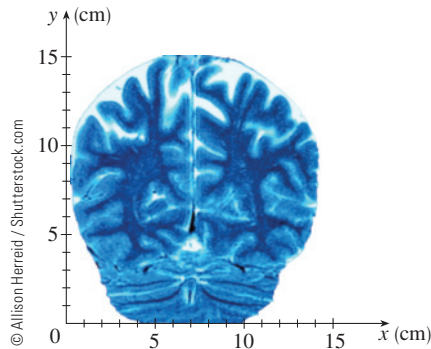
Answers to the Concept Check can be found on the back endpapers.

### EXERCISES

**1–4** Find the area of the region bounded by the given curves.

- 1.**  $y = x^2$ ,  $y = 4x - x^2$
- 2.**  $y = 1/x$ ,  $y = x^2$ ,  $y = 0$ ,  $x = e$
- 3.**  $y = 1 - 2x^2$ ,  $y = |x|$
- 4.**  $x + y = 0$ ,  $x = y^2 + 3y$

- 5. MRI brain scan** Shown is a cross-section of a human brain obtained with an MRI. Use the Midpoint Rule to estimate the area of the cross-section.



- 6. Birth and death rates** The birth rate of a population is  $b(t) = 1240e^{0.0197t}$  people per year and the death rate is  $d(t) = 682e^{0.008t}$  people per year. Find the area between these curves for  $0 \leq t \leq 20$ . What does this area represent?

- 7.** Find the average value of the function  $f(t) = t \sin(t^2)$  on the interval  $[0, 10]$ .
- 8.** Find the average value of the function  $f(x) = x^2\sqrt{1+x^3}$  on the interval  $[0, 2]$ .

- 9. Antibiotic pharmacokinetics** When an antibiotic tablet is taken, the concentration of the antibiotic in the bloodstream is modeled by the function

$$C(t) = 8(e^{-0.4t} - e^{-0.6t})$$

where the time  $t$  is measured in hours and  $C$  is measured in  $\mu\text{g/mL}$ . What is the average concentration of the antibiotic during the first two hours?

- 10. Salicylic acid pharmacokinetics** In a study of the effects of aspirin, salicylic acid was formed and its concentration was modeled by the function

$$C(t) = 11.4te^{-t}$$

where the time  $t$  is measured in hours and  $C$  is measured in  $\mu\text{g/mL}$ . What is the average concentration of the salicylic acid during the first four hours?

- 11. Survival and renewal** Suppose a city's population is currently 75,000 and the renewal function is

$$R(t) = 3200e^{0.05t}$$

If the survival function is  $S(t) = e^{-0.1t}$ , predict the population in 10 years.

- 12. Animal survival and renewal** The fish population in a lake is currently 3400 and is increasing at a rate of  $R(t) = 650e^{0.04t}$  fish per month. If the proportion of fish

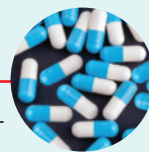
that remain after  $t$  months is  $S(t) = e^{-0.09t}$ , how many fish will be in the lake in three years?

- 13. Cardiac output** After a 6-mg injection of dye into a heart, the readings of dye concentration, in  $\text{mg/L}$ , at two-second intervals are as shown in the table. Use the Midpoint Rule to estimate the cardiac output.

| $t$ | $C(t)$ | $t$ | $C(t)$ |
|-----|--------|-----|--------|
| 0   | 0      | 14  | 4.7    |
| 2   | 1.9    | 16  | 3.3    |
| 4   | 3.3    | 18  | 2.1    |
| 6   | 5.1    | 20  | 1.1    |
| 8   | 7.6    | 22  | 0.5    |
| 10  | 7.1    | 24  | 0      |
| 12  | 5.8    |     |        |

- 14.** Find the volume of the solid obtained by rotating about the  $x$ -axis the region bounded by the curves  $y = e^{-2x}$ ,  $y = 1 + x$ , and  $x = 1$ .
- 15.** Let  $\mathcal{R}$  be the region bounded by the curves  $y = \tan(x^2)$ ,  $x = 1$ , and  $y = 0$ . Use the Midpoint Rule with  $n = 4$  to estimate the following quantities.
- The area of  $\mathcal{R}$
  - The volume obtained by rotating  $\mathcal{R}$  about the  $x$ -axis
- 16.** Let  $\mathcal{R}$  be the region in the first quadrant bounded by the curves  $y = x^3$  and  $y = 2x - x^2$ . Calculate the following quantities.
- The area of  $\mathcal{R}$
  - The volume obtained by rotating  $\mathcal{R}$  about the  $x$ -axis
- 17.** Find the volumes of the solids obtained by rotating the region bounded by the curves  $y = x$  and  $y = x^2$  about the following lines.
- The  $x$ -axis
  - The  $y$ -axis
-  **18.** Let  $\mathcal{R}$  be the region bounded by the curves  $y = 1 - x^2$  and  $y = x^6 - x + 1$ . Estimate the following quantities.
- The  $x$ -coordinates of the points of intersection of the curves
  - The area of  $\mathcal{R}$
  - The volume generated when  $\mathcal{R}$  is rotated about the  $x$ -axis
- 19.** The base of a solid is a circular disk with radius 3. Find the volume of the solid if parallel cross-sections perpendicular to the base are isosceles right triangles with hypotenuse lying along the base.
- 20.** The height of a monument is 20 m. A horizontal cross-section at a distance  $x$  meters from the top is an equilateral triangle with side  $\frac{1}{4}x$  meters. Find the volume of the monument.

## CASE STUDY 1d Kill Curves and Antibiotic Effectiveness



In this case study we have explored the relationship between the magnitude of antibiotic treatment and the effectiveness of the treatment. To do so, in Case Study 1b we showed that a suitable model for the size of the bacteria population  $P(t)$  (in CFU/mL) as a function of time  $t$  (in hours) is given by the piecewise defined function

$$(1a) \quad P(t) = \begin{cases} 6e^{t/3} & \text{if } t < 2.08 \\ 12 & \text{if } t \geq 2.08 \end{cases}$$

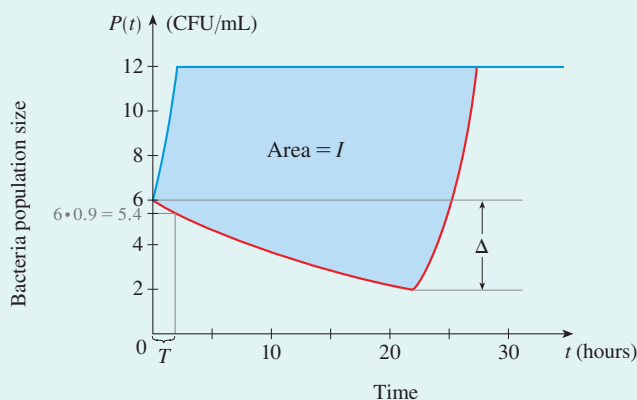
Recall that  $MIC$  is a constant referred to as the minimum inhibitory concentration of the antibiotic and  $c_0$  is the concentration at time  $t = 0$ .

if  $c_0 < MIC$ , where  $MIC = 0.013 \mu\text{g/mL}$ . On the other hand, if  $c_0 > MIC$

$$(1b) \quad P(t) = \begin{cases} 6e^{-t/20} & \text{if } t < a \\ 6Ae^{t/3} & \text{if } a \leq t < b \\ 12 & \text{if } t \geq b \end{cases}$$

where the constants  $a$ ,  $b$ , and  $A$  are defined by  $a = 5.7 \ln(77c_0)$ ,  $b = 6.6 \ln(77c_0) + 2.08$ , and  $A = (77c_0)^{-2.2}$ .

In Case Study 1a we used these so-called kill curves to plot the relationship between  $\alpha$  (a measure of the magnitude of antibiotic treatment) and two different measures of the killing effectiveness, denoted by  $\Delta$  and  $T$ . The quantity  $\Delta$  is the drop in population size before the population rebound occurs, and  $T$  is the time taken to reduce the bacteria population size to 90% of its initial size. (Refer to Figure 1.)



**FIGURE 1**

Three measures of killing effectiveness. The blue curve is bacteria population size in the absence of antibiotic. The red curve is bacteria population size in the presence of antibiotic.

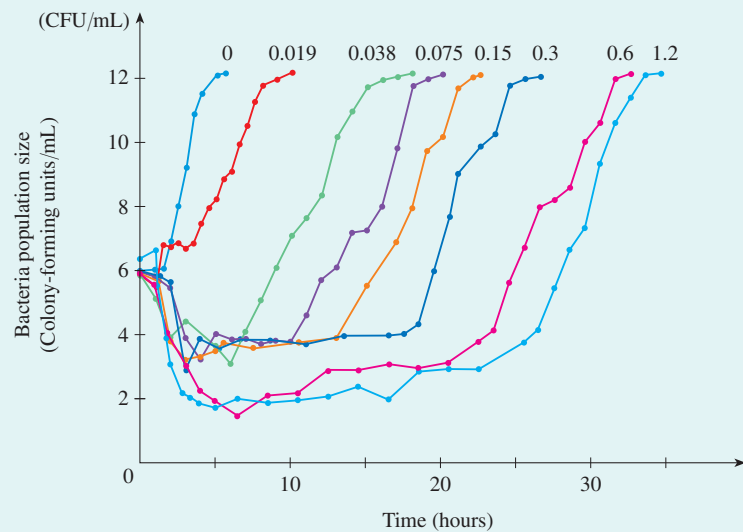
Now that we have a model that works reasonably well, we can use it to make predictions about other patterns, and then compare these with available data. As an example, another measure of the killing effectiveness of an antibiotic is the area  $I$  between the population size curve in the absence of antibiotic, and the kill curve in the presence of the antibiotic as shown in Figure 1. In many cases this measure might be preferable because it incorporates both the drop in bacteria population size, and the length of time for which this reduced population size is maintained. Let's see what our model predicts about the relationship between  $I$  and the magnitude of antibiotic treatment  $\alpha$ .

1. Suppose that  $a > 2.08$  [that is,  $5.7 \ln(77c_0) > 2.08$ ]. Find an expression for  $I$  in the modeled populations in terms of  $c_0$ . You should assume that  $c_0 > MIC$ .

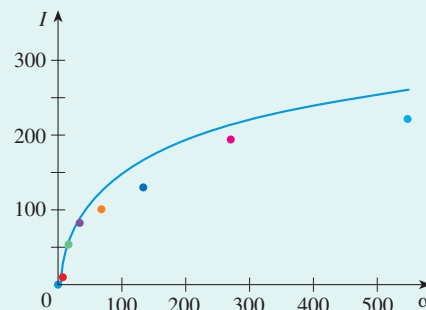
2. The result from Problem 1 should give you a function of the form  $I = g(c_0)$  for some function  $g$ . Substitute the values  $k = 0.175$  and  $MIC = 0.013$  into the expression for  $\alpha$  obtained in Case Study 1c. This will give  $\alpha = h(c_0)$  for some function  $h$ .
3. Using the concept of a function's inverse, explain how to obtain an expression giving  $I$  as a function of  $\alpha$  in terms of  $g$  and  $h^{-1}$ . Find an explicit expression for this function.
4. Plot the function obtained in Problem 3.

The experimental kill curves shown in Figure 2 have also been used to quantify the relationship between  $I$  and  $\alpha$ .<sup>1</sup> In other words, the values of  $I$  and  $\alpha$  have been calculated for each experimental kill curve in Figure 2. If we overlay these data points on the plot from Problem 4, we obtain Figure 3. You can see that, again, our model predicts the observed data reasonably well.

**FIGURE 2**  
The kill curves of ciprofloxacin for *E. coli* when measured in a growth chamber. The concentration of ciprofloxacin at  $t = 0$  is indicated above each curve (in  $\mu\text{g/mL}$ ).



**FIGURE 3**  
Predicted relationship between  $I$  and  $\alpha$ , along with experimental observations obtained using the kill curve data in Figure 2.



1. Adapted from A. Firsov et al., "Parameters of Bacterial Killing and Regrowth Kinetics and Antimicrobial Effect in Terms of Area Under the Concentration-Time Curve Relationships: Action of Ciprofloxacin against *Escherichia coli* in an In Vitro Dynamic Model." *Antimicrobial Agents and Chemotherapy* 41 (1997): 1281.

## CASE STUDY 2b Hosts, Parasites, and Time-Travel



In Case Study 2c you will derive a model for the dynamics of the genotypes of *Daphnia* and its parasite. Recall that we are modeling a situation involving two possible host genotypes (A and a) and two possible parasite genotypes (B and b). Parasites of type B can infect only hosts of type A, while parasites of type b can infect only hosts of type a. You will then derive an explicit solution of a simplified version of the model in Case Study 2d. This will give the frequency of host genotype A and parasite genotype B as functions of time. These functions are

$$(1) \quad \begin{aligned} q(t) &= \frac{1}{2} + M_q \cos(ct - \phi_q) \\ p(t) &= \frac{1}{2} + M_p \cos(ct - \phi_p) \end{aligned}$$

where  $q(t)$  is the predicted frequency of host genotype A at time  $t$  and  $p(t)$  is the predicted frequency of the parasite genotype B at time  $t$ . In these equations  $\phi_q$ ,  $\phi_p$ , and  $c$  are positive constants, and  $M_q$  and  $M_p$  are positive constants that are strictly less than  $\frac{1}{2}$  (the biological significance of these constants is explored in Case Study 2a).

In this part of the case study you will use Equations 1 to make predictions from the model that can be compared with data from the experiments.

Recall that, in the experiment, a host from a fixed layer of the sediment core was challenged with infection by a parasite from either the same layer, a layer above the fixed layer (that is, from its future), or a layer below it (that is, from its past). We can view different depths in the sediment core as representing different points of time in the history of the *Daphnia*-parasite interaction (see Figure 1). In this way Equations 1 can equally be viewed as specifying the frequency of the host and parasite genotypes as functions of location in the sediment core. Increasing values of  $t$  correspond to shallower points in the core as shown in Figure 2.

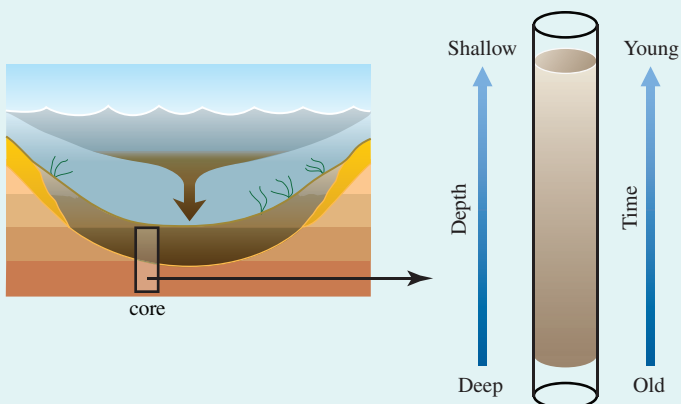


FIGURE 1

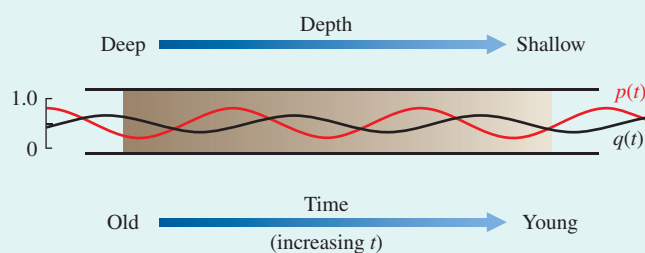


FIGURE 2

In the experiment introduced in Case Study 2 on page xlvii, researchers chose a fixed depth  $\tau$  and extracted a layer of sediment of width  $W$  centered around this depth. This layer is shown in Figure 3.

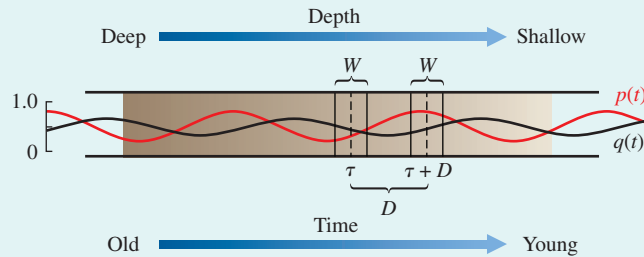


FIGURE 3

After the contents of this layer were completely mixed, hosts and parasites were extracted at random from the mixture. Researchers also took deeper and shallower layers (that represent the past and the future for hosts located in the layer at  $\tau$ ) and completely mixed each. The center of these layers was a distance  $D$  from the center of the focal layer (see Figure 3). They then challenged hosts from the layer at  $\tau$  with parasites from their past (that is, from the layer with  $D < 0$ ), present (the layer at  $\tau$ ), and future (the layer with  $D > 0$ ). For each challenge experiment the fraction of hosts becoming infected was measured.

We can use our model to predict the fraction of hosts infected. To do so, we first need to know the predicted frequency of hosts of type A in the layer at  $\tau$  as well as the frequency of the parasites of type B in the layer at  $\tau + D$ .

1. Consider a focal layer at location  $\tau$  with width  $W$  as shown in Figure 4.

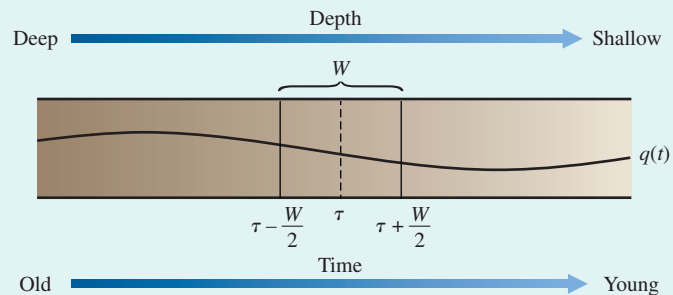


FIGURE 4

The frequency of host type A will vary across the depth of this layer as specified by the function  $q(t)$ . Show that, when this layer is completely mixed, the frequency of A in the mixture is given by

$$q_{\text{ave}}(\tau) = \frac{1}{2} + M_q \cos(c\tau - \phi_q) \frac{2 \sin(\frac{1}{2}cW)}{cW}$$

*Hint:* You might want to use the trigonometric identity

$$\sin(x + y) - \sin(x - y) = 2 \cos x \sin y$$

2. Consider another layer at a location a distance  $D$  from  $\tau$ , again with thickness  $W$ , as shown in Figure 5. The frequency of parasite type B will vary across depth in this layer, as specified by the function  $p(t)$ . Show that, when this layer is completely mixed, the frequency of B in the mixture is given by

$$p_{\text{ave}}(\tau + D) = \frac{1}{2} + M_p \cos(c(\tau + D) - \phi_p) \frac{2 \sin(\frac{1}{2}cW)}{cW}$$

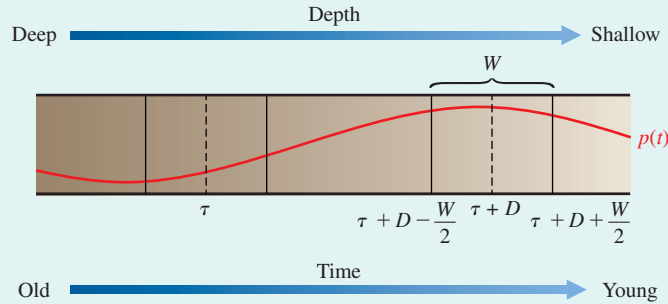


FIGURE 5

3. Suppose that hosts from the layer at  $\tau$  are challenged with parasites from the layer at  $\tau + D$ . Use the facts that only B parasites can infect A hosts and only b parasites can infect a hosts to explain why the fraction of hosts infected in this challenge experiment is predicted to be

$$I(\tau) = p_{\text{ave}}(\tau + D)q_{\text{ave}}(\tau) + [1 - p_{\text{ave}}(\tau + D)][1 - q_{\text{ave}}(\tau)]$$

The final step is to recognize that the experiment was actually conducted with several different, randomly chosen depths  $\tau$ . Therefore we need to average  $I(\tau)$  in Problem 3 over the possible depths. Because  $I(\tau)$  is periodic, we need only average over one period of its cycle. Its average is therefore

$$F = \frac{1}{T} \int_0^T I(\tau) d\tau$$

where  $T = 2\pi/c$  is the period of  $I(\tau)$ .

4. Show that

$$(2) \quad F(D) = \frac{1}{2} + M_p M_q \cos(cD - \phi^*) \frac{4 \sin^2(\frac{1}{2}cW)}{c^2 W^2}$$

where  $\phi^* = \phi_p - \phi_q$ .

*Hint:* You might want to use the trigonometric identity

$$\cos x \cos y = \frac{1}{2}(\cos(x + y) + \cos(x - y))$$

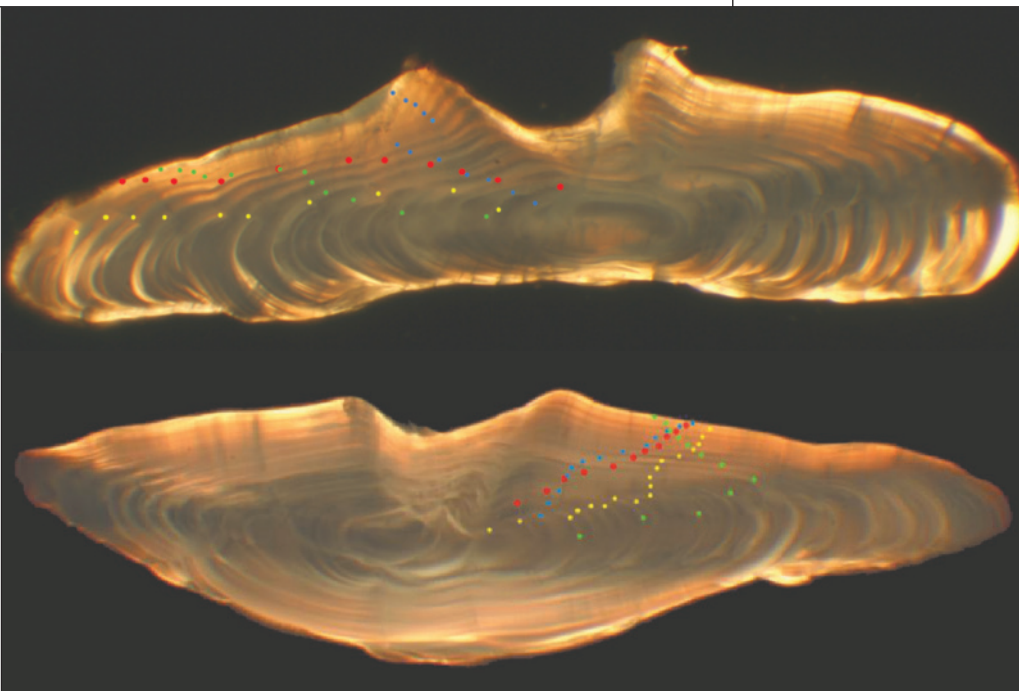
Equation 2 was used in Case Study 2a to predict the experimental outcome expected under different conditions.

# Differential Equations

# 7

Shown are otoliths from Atlantic redfish—they were used to estimate fish age when fitting the von Bertalanffy differential equation in Example 7.4.2.

Dr. Cristoph Stransky/Thuenen Institute of Sea Fisheries



## 7.1 Modeling with Differential Equations

PROJECT: Chaotic Blowflies and the Dynamics of Populations

## 7.2 Phase Plots, Equilibria, and Stability

PROJECT: Catastrophic Population Collapse: An Introduction to Bifurcation Theory

## 7.3 Direction Fields and Euler's Method

## 7.4 Separable Equations

PROJECT: Why Does Urea Concentration Rebound After Dialysis?

## 7.5 Systems of Differential Equations

PROJECT: The Flight Path of Hunting Raptors

## 7.6 Phase Plane Analysis

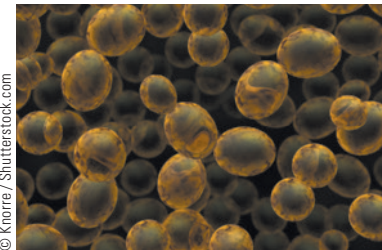
PROJECT: Determining the Critical Vaccination Coverage

CASE STUDY 2c: Hosts, Parasites, and Time-Travel

ONE OF THE MOST IMPORTANT applications of calculus is to differential equations. A wide variety of biological processes can be modeled using differential equations, and such equations have provided enormous insight into our understanding of the dynamics of living organisms—how individuals and populations change over time.

7.1     Modeling with Differential Equations

Many biological processes occur continuously through time. Examples include the change in concentration of a drug in the bloodstream of a patient, or the growth in mass of individual organisms. Even the population dynamics of many species, from size of bacteria colonies to the size of the human population, are sometimes best modeled by assuming the quantity of interest (population size, in this case) changes continuously through time. (For example, see page 146.) As we will see in this chapter, differential equations provide a convenient and natural way to construct such models.



Models of Population Growth

A **differential equation** is an equation that contains an unknown function and one or more of its derivatives. Such equations arise in a variety of situations but one of the most common is in models of population growth.

Consider the growth of a population of yeast. Yeast are single-celled organisms used for a variety of purposes, including alcohol production and baking. Researchers collected the data in Table 1 from a yeast population grown in liquid culture, measuring the population size (in number of individuals per mL of culture) at different points in time (in hours).<sup>1</sup> Figure 1 is a scatter plot of these data.

Table 1

| Time (h) | Pop. size ( × 10 <sup>6</sup> /mL) | Time (h) | Pop. size ( × 10 <sup>6</sup> /mL) |
|----------|------------------------------------|----------|------------------------------------|
| 0        | 0.200                              | 19       | 209                                |
| 1        | 0.330                              | 20       | 190                                |
| 2        | 0.500                              | 21       | 210                                |
| 3        | 1.10                               | 22       | 200                                |
| 4        | 1.40                               | 23       | 215                                |
| 5        | 3.10                               | 24       | 220                                |
| 6        | 3.50                               | 25       | 200                                |
| 7        | 9.00                               | 26       | 180                                |
| 8        | 10.0                               | 27       | 213                                |
| 9        | 25.4                               | 28       | 210                                |
| 10       | 27.0                               | 29       | 210                                |
| 11       | 55.0                               | 30       | 220                                |
| 12       | 76.0                               | 31       | 213                                |
| 13       | 115                                | 32       | 200                                |
| 14       | 160                                | 33       | 211                                |
| 15       | 162                                | 34       | 200                                |
| 16       | 190                                | 35       | 208                                |
| 17       | 193                                | 36       | 230                                |
| 18       | 190                                |          |                                    |

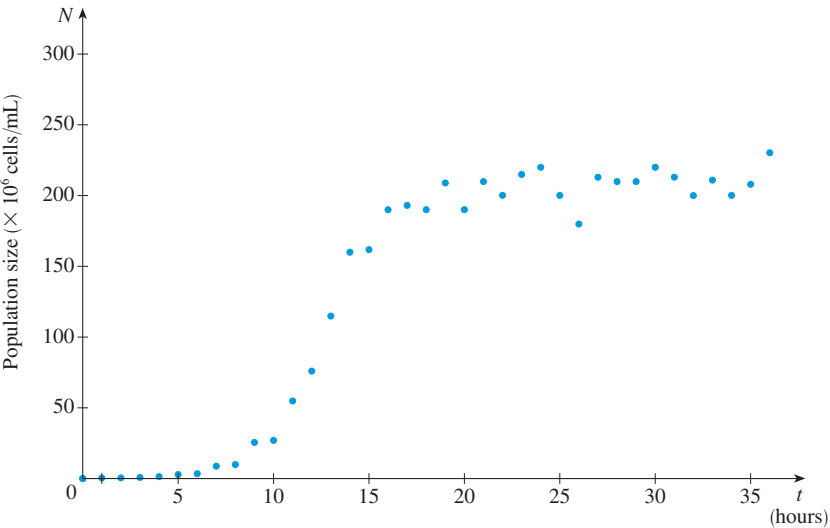


FIGURE 1     A scatter plot of the data in Table 1

1. B. K. Mable et al., “Masking and Purging Mutations following EMS Treatment in Haploid, Diploid, and Tetraploid Yeast (*Saccharomyces cerevisiae*),” *Genetical Research* 77 (2001): 9–26.

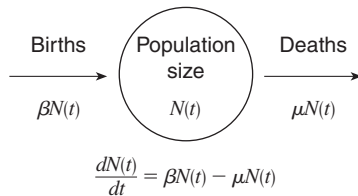


FIGURE 2

Equation 3 can be derived directly by simply *assuming* that the yeast population grows at a rate proportional to its size. The rate of growth of the population is the derivative  $dN/dt$ , and therefore we obtain Equation 3, where  $r$  is a constant of proportionality. See Equation 3.6.1, where it was called the *law of natural growth*.

BB

Although the population size in Table 1 was measured at one-hour intervals, the yeast themselves are replicating in a way that is nearly continuous in time. In other words, no matter how small we make the interval of time between successive measurements, some reproduction and death will likely have occurred.

How can we model such processes? Let's start simply and assume that each individual yeast cell produces offspring at a constant rate  $\beta$ . Thus the total rate of offspring production (that is, the total birth rate) at time  $t$  is  $\beta N(t)$ , where  $N(t)$  is the number of yeast cells present at time  $t$ . Likewise, suppose the total loss rate of yeast cells through death at time  $t$  is  $\mu N(t)$ , where  $\mu$  is a constant death rate per individual cell.

With the preceding assumptions, we see that the rate of change of the number of yeast cells at time  $t$  is the total birth rate minus the total death rate,  $\beta N(t) - \mu N(t)$ . And since the rate of change of  $N(t)$ , the number of yeast cells, can also be written as  $dN(t)/dt$ , we can write

$$(1) \quad \frac{dN(t)}{dt} = \beta N(t) - \mu N(t)$$

(See Figure 2.) Now if we define the constant  $r$  as

$$(2) \quad r = \beta - \mu$$

then Equation 1 can be written more simply as

$$(3) \quad \frac{dN(t)}{dt} = rN(t)$$

The quantity  $r$  in Equation 2 is called the **per capita growth rate**. It is the rate of growth of the population *per individual* in the population. Since  $dN/dt$  is the rate of growth of the population, the rate of growth *per individual* is  $dN/dt$  divided by  $N(t)$ . From Equation 3, we get

$$\frac{dN(t)}{dt} \frac{1}{N(t)} = r$$

showing that  $r$  is indeed the per capita growth rate.

Equation 3 involves the unknown function  $N(t)$  along with its first derivative and is therefore a differential equation. The population size  $N$  is the *dependent variable* and time  $t$  is the *independent variable*. This differential equation tells us that the rate of change of the population size of yeast at any time is proportional to the size of the population at that time. Put another way, the rate of reproduction of each individual in the population (that is, the *per capita* rate of reproduction) is constant and equal to  $r$ .

The model given by Equation 3 is one of the simplest models for population growth. Let's see how well it predicts the data in Table 1. First notice that if  $r > 0$ , then from Equation 3

$$\frac{dN(t)}{dt} = rN(t) > 0$$

Biologically, if the per capita growth rate is positive (meaning that the birth rate  $\beta$  is larger than the death rate  $\mu$ ), then the yeast population will increase. On the other hand, if  $r < 0$  (the birth rate  $\beta$  is smaller than the death rate  $\mu$ ), then from Equation 3

$$\frac{dN(t)}{dt} = rN(t) < 0$$

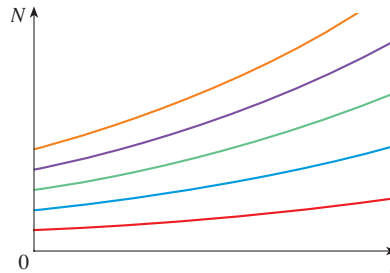
and the yeast population will decrease.

To make more progress, we would like to obtain an explicit function  $N(t)$  that tells us exactly what the population size will be at any time. Such a function  $N(t)$  is called a *solution* of the differential equation. It is a function that, when substituted into both sides of the differential equation, produces an equality.

Equation 3 tells us that  $N(t)$  is a function whose derivative is equal to the function itself, multiplied by a constant,  $r$ . As we have seen in Chapter 3, exponential functions have exactly this property. In fact we can see that the function  $N(t) = Ce^{rt}$  satisfies the differential equation. In particular, substituting this choice of  $N(t)$  into Equation 3, we obtain

$$N'(t) = C(re^{rt}) = r(Ce^{rt}) = rN(t)$$

demonstrating that  $N(t) = Ce^{rt}$  does, in fact, satisfy the differential equation. (We will see in Section 7.4 that there is no other solution.) Here  $C$  is an arbitrary constant. We can obtain a biological interpretation of this constant by setting  $t = 0$ : This gives  $N(0) = Ce^{r(0)} = C$ , revealing that  $C$  is the population size at  $t = 0$ . Figure 3 shows examples of the solution curves for different values of  $C$  when  $r > 0$ .



**FIGURE 3**

The family of solutions  $N(t) = Ce^{rt}$  with  $r > 0$ ,  $t \geq 0$ , and different values of  $C$

We can already see from Figure 3 that Equation 3 does not capture all of the features of the data in Figure 1. For example, it appears to predict continued population growth. To obtain a more satisfying comparison, however, we should choose appropriate values for the constants  $C$  and  $r$ .

From the data in Table 1 we see that  $N(0) = 0.200$  and therefore  $C = 0.200$ . One way to obtain a suitable value for  $r$  is to consider the factor by which the population of yeast grew over some fixed period of time. For example, in the first hour the yeast population grew by a factor of

$$\frac{0.330}{0.200} = 1.65$$

On the other hand, according to the model, the factor by which this population is predicted to have grown is

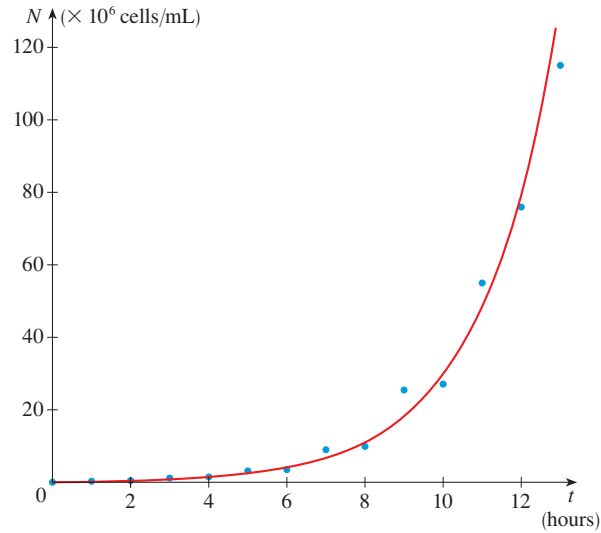
$$\frac{N(1)}{N(0)} = \frac{Ce^{r \cdot 1}}{Ce^{r \cdot 0}} = e^r$$

Therefore a reasonable choice for  $r$  would be the value for which  $e^r = 1.65$ . Solving this equation for  $r$  gives  $r = \ln 1.65 \approx 0.5$ . Thus, our final model is

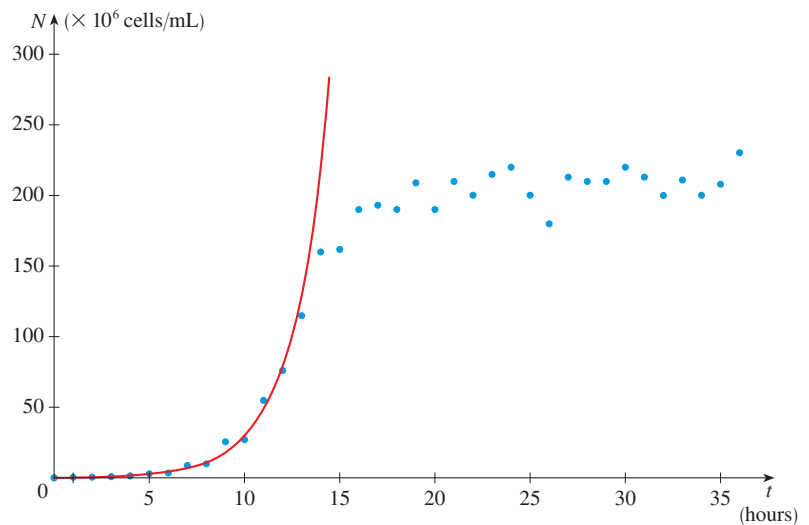
$$N(t) = 0.2e^{0.5t}$$

Figures 4(a) and 4(b) both plot this equation along with the data from Table 1, but on two different intervals of time. The model provides remarkably accurate predictions

over the first 13 or so hours, as shown in Figure 4(a), but its predictions are extremely inaccurate for later time points in the data [see Figure 4(b)].



(a)



(b)

**FIGURE 4**

In retrospect, one obvious biological reason for this discrepancy is that the model assumes the per capita growth rate remains constant at  $r$ , regardless of the population size. In reality, as the population gets large, we might expect that crowding and resource depletion will cause the per capita growth rate to decline.

**BB**

In fact, using the data in Table 1, it is possible to show that the per capita growth rate for the yeast population varies as a function of population size according to the equation

$$\text{per capita growth rate} \approx 0.55 - 0.0026N$$

See Exercise 11.3.25.

In other words,

$$\frac{dN(t)}{dt} \frac{1}{N(t)} \approx 0.55 - 0.0026N(t)$$

Thus a better differential equation for modeling the yeast population is

$$\frac{dN}{dt} = (0.55 - 0.0026N)N$$

We will learn how to analyze differential equations of this form in later sections. For now we simply note that these techniques can be used to show that the solution is

$$N(t) = \frac{42e^{0.55t}}{209.8 + 0.2e^{0.55t}}$$

(See Exercise 18.) This function is plotted in Figure 5 along with the data from Table 1. We see that this model provides quite accurate predictions over the entire time period of the experiment.

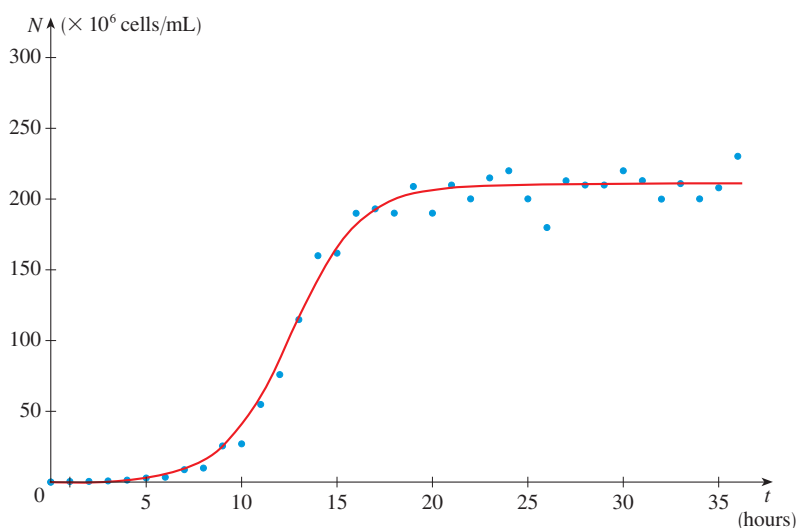


FIGURE 5

Our revised yeast model is a specific example of a more general model for population growth called the *logistic differential equation*. Suppose that the per capita growth rate of a population decreases linearly as the population size increases, from a value of  $r$  when  $N = 0$  to a value of 0 when  $N = K$ . The positive constant  $K$  is referred to as the *carrying capacity*; it is the population size at which crowding and resource depletion cause the per capita growth rate to be zero. In Exercise 16 you are asked to show that this results in the differential equation

$$(4) \quad \frac{dN}{dt} = r \left( 1 - \frac{N}{K} \right) N$$

The logistic growth equation was first proposed by Dutch mathematical biologist Pierre-François Verhulst in the 1840s as a model for world population growth.

Equation 4 is called the **logistic differential equation**, or more simply the **logistic equation**. In Exercise 17 you are asked to show that, for the yeast model,  $r = 0.55$  and  $K \approx 210$ .

We can obtain some qualitative features of the solutions of Equation 4 by inspection. We first observe that the constant functions  $N(t) = 0$  and  $N(t) = K$  are solutions because, in either case, the left side of Equation 4 is then zero (the derivative of a constant is zero), and the right side is zero as well. Such constant solutions are called *equilibrium solutions*. (A formal definition of an equilibrium solution will be given in Section 7.2.)

If the initial population  $N(0)$  lies between 0 and  $K$ , then the right side of Equation 4 is positive, so  $N'(t) > 0$  and the population increases (assuming  $r > 0$ ). But if the population exceeds the carrying capacity ( $N > K$ ), then  $1 - N/K$  is negative, so  $N'(t) < 0$  and the population decreases. In either case, if the population approaches the carrying capacity ( $N \rightarrow K$ ), then  $N'(t) \rightarrow 0$ , which means the population levels off.

## ■ Classifying Differential Equations

Differential equations involve an unknown function and its derivatives. The **order** of the differential equation is the order of the highest derivative appearing in the equation. For example,  $y'(t) + 2y(t) = 3$  is a first-order differential equation, whereas  $5y''(t) - y'(t) = y(t)$  is a second-order differential equation. The **solution** of a differential equation is a function that, when substituted into the equation, produces an equality. For example, we can verify that the function  $y(t) = e^t - 2$  is a solution of the differential equation  $dy/dt = 2 + y(t)$  as follows: Substituting the function into the left side of this differential equation gives

$$\frac{dy}{dt} = \frac{d}{dt}(e^t - 2) = e^t$$

and substituting it into the right side gives

$$2 + (e^t - 2) = e^t$$

The right and left sides evaluate to the same expression, demonstrating that the function  $y(t) = e^t - 2$  is indeed a solution.

Typically, there are several solutions to a differential equation. In many problems we need to find the particular solution that satisfies an additional condition of the form  $y(t_0) = y_0$ . This is called an **initial condition**. The problem of finding a solution of the differential equation that also satisfies an initial condition is called an **initial-value problem**. Graphically, when we impose an initial condition, we look at the family of solution curves and pick the one that passes through the point  $(t_0, y_0)$ . For an example involving the logistic equation, see Figure 6. This corresponds to measuring the state of a system at time  $t_0$  and using the solution of the initial-value problem to predict the future behavior of the system.

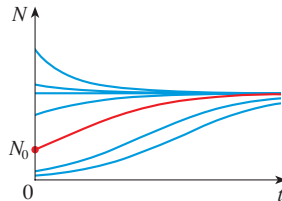
Verifying a solution is relatively easy, but obtaining a solution in the first place may not be. The difficulty of this task—and indeed whether or not it is even possible—is determined by the type of the differential equation. We consider three types of first-order differential equations: pure time, autonomous, and nonautonomous differential equations.

## Pure-Time Differential Equations

**Pure-time differential equations** involve the derivative of the function but not the function itself. For example, if the rate of change of population size  $y$  depends on time only, this results in a differential equation of the form

$$\frac{dy}{dt} = f(t)$$

We have already studied this type of equation in the context of antidifferentiation (in Section 4.6) and integration (in Chapter 5). We can obtain the solution  $y(t)$  by calculating the antiderivative of  $f(t)$ . Although we refer to such equations as *pure-time differential equations*, the independent variable need not be time.



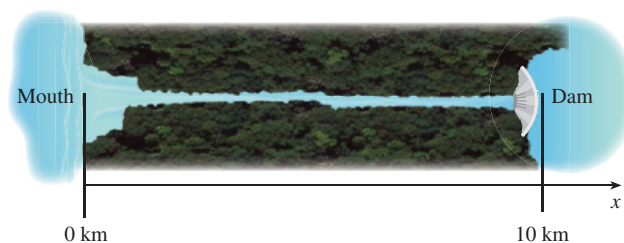
**FIGURE 6**

The family of solutions of the logistic equation. The solution curve satisfying the initial condition  $N(0) = N_0$  is shown in red.

**EXAMPLE 1** | Spatial species distributions As we move up a stream from its mouth toward its source, suppose that the population size  $n$  of a species of insect at a fixed point in time changes over space according to

$$\frac{dn}{dx} = 1 - 2e^{-x}$$

where  $0 \leq x \leq 10$  is the spatial location (in km) between the mouth ( $x = 0$  km) and a dam ( $x = 10$  km). (This situation is described in Figure 7.) Suppose the population size at the dam is  $n(10) = 20$ . Obtain an expression for the population size as a function of distance from the mouth.



**FIGURE 7**

Population density along a stream

**SOLUTION** We first seek a function  $n(x)$  that satisfies the given differential equation. This function can be obtained by integrating both sides of the differential equation with respect to  $x$ :

$$\frac{dn}{dx} = 1 - 2e^{-x}$$

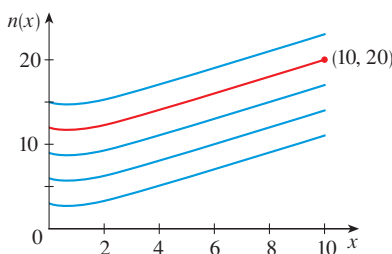
$$\int \frac{dn}{dx} dx = \int (1 - 2e^{-x}) dx$$

$$n(x) = x + 2e^{-x} + C$$

The function  $n(x) = x + 2e^{-x} + C$  is a family of solutions. We now need to choose the specific function from this family that satisfies the condition  $n(10) = 20$ . Substituting  $x = 10$  into  $n(x)$  gives

$$n(10) = 10 + 2e^{-10} + C = 20$$

This tells us that we must choose  $C = 10 - 2e^{-10}$ . Therefore the population size as a function of  $x$  is  $n(x) = x + 2e^{-x} + 10 - 2e^{-10}$ . See Figure 8.



**FIGURE 8**

The family of solutions giving population size along the stream. The solution curve satisfying the initial condition  $n(10) = 20$  is shown in red.

## Autonomous Differential Equations

**Autonomous differential equations** arise when the equation involves both the derivative of the function and the function itself, but when there is no explicit dependence on the independent variable. Such equations have the general form

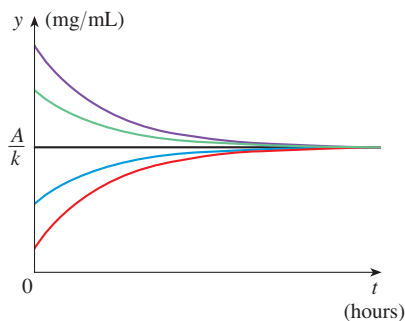
$$\frac{dy}{dt} = g(y)$$

where  $y$  is the unknown function of the independent variable  $t$ . Equations 3 and 4 are examples of autonomous differential equations.

**EXAMPLE 2** | **BB** Modeling intravenous drug delivery Often the rate at which the body metabolizes a drug is proportional to the current concentration of the drug. In other words, if  $y(t)$  is the concentration of a drug in the bloodstream at time  $t$



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**FIGURE 9**

The family of solutions of Equation 5

(measured in mg/mL), then

$$\text{outflow of drug through metabolism} = ky$$

where  $k$  is a positive constant of proportionality (with units 1/hour).

For drugs administered through a constant intravenous supply, the concentration in the bloodstream will also be replenished at a rate that is determined by the drug concentration in the supply:

$$\text{inflow of drug through IV supply} = A$$

where  $A$  is a positive constant with units mg/(mL hour). The total rate of change of concentration resulting from both processes (that is,  $dy/dt$ ) is therefore

$$\frac{dy}{dt} = \text{inflow} - \text{outflow}$$

or

$$(5) \quad \frac{dy}{dt} = A - ky$$

Equation 5 is an autonomous differential equation because it involves the dependent variable  $y$  but not the independent variable  $t$ . Figure 9 shows a family of solutions to differential equation (5), and suggests that the drug concentration is predicted to approach a limiting value of  $A/k$  at time passes, regardless of the initial concentration.

## Nonautonomous Differential Equations

**Nonautonomous differential equations** are a combination of pure-time and autonomous differential equations. They arise when the equation involves the function and its derivative, and the independent variable appears explicitly as well.

**EXAMPLE 3 | Administering drugs** A drug is administered to a patient intravenously at a time-varying rate of  $A(t) = 1 + \sin t$  mg/(mL hour), and is metabolized at a rate of  $y(t)$  mg/(mL hour), where  $y(t)$  is the concentration at time  $t$  (in units of mg/mL). Thus  $y$  obeys the differential equation

$$(6) \quad \frac{dy}{dt} = 1 + \sin t - y$$

Verify that the family of functions  $y(t) = Ce^{-t} + \frac{1}{2}(2 - \cos t + \sin t)$  satisfies the differential equation.

**SOLUTION** Substituting  $y(t)$  into the left side of the differential equation (6) gives  $-Ce^{-t} + \frac{1}{2}(\sin t + \cos t)$ . Substituting it into the right side gives

$$\begin{aligned} 1 + \sin t - y &= 1 + \sin t - Ce^{-t} - \frac{1}{2}(2 - \cos t + \sin t) \\ &= -Ce^{-t} + \frac{1}{2}(\sin t + \cos t) \end{aligned}$$

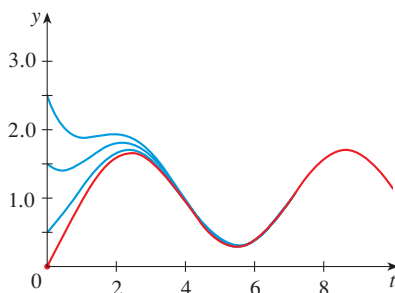
Since both quantities are the same, this family of functions  $y$  satisfies the differential equation.

**EXAMPLE 4 | Administering drugs (continued)** What is the drug concentration as a function of time for the model in Example 3 if the initial drug concentration at  $t = 0$  is zero?

**SOLUTION** We seek the specific member from the family of functions,  $y(t) = Ce^{-t} + \frac{1}{2}(2 - \cos t + \sin t)$ , that also satisfies  $y(0) = 0$ . Evaluating, we obtain

$$y(0) = Ce^{-0} + \frac{1}{2}(2 - \cos 0 + \sin 0) = C + \frac{1}{2} = 0$$

and therefore  $C = -\frac{1}{2}$ . Thus the solution to the *initial-value problem* is  $y(t) = \frac{1}{2}(2 - e^{-t} - \cos t + \sin t)$ , as shown in Figure 10.



**FIGURE 10**

The family of solutions giving drug concentration  $y(t)$ . The solution curve satisfying the initial condition  $y(0) = 0$  is shown in red.

## EXERCISES 7.1

1. Show that  $y = \frac{2}{3}e^x + e^{-2x}$  is a solution of the differential equation  $y' + 2y = 2e^x$ . Is this differential equation pure-time, autonomous, or nonautonomous?
2. Verify that  $y = -t \cos t - t$  is a solution of the initial-value problem

$$t \frac{dy}{dt} = y + t^2 \sin t \quad y(\pi) = 0$$

Is this differential equation pure-time, autonomous, or nonautonomous?

3. Show that  $y = e^{-at} \cos t$  is a solution of the differential equation  $y' = -e^{-at}(a \cos t + \sin t)$ . Is this differential equation pure-time, autonomous, or nonautonomous?
4. (a) Show that every member of the family of functions  $y = (\ln x + C)/x$  is a solution of the differential equation  $x^2 y' + xy = 1$ .  
(b) Illustrate part (a) by graphing several members of the family of solutions on a common screen.  
(c) Find a solution of the differential equation that satisfies the initial condition  $y(1) = 2$ .  
(d) Find a solution of the differential equation that satisfies the initial condition  $y(2) = 1$ .
5. (a) What can you say about a solution of the equation  $y' = -y^2$  just by looking at the differential equation?  
(b) Verify that all members of the family  $y = 1/(x + C)$  are solutions of the equation in part (a).

- (c) Can you think of a solution of the differential equation  $y' = -y^2$  that is not a member of the family in part (b)?
- (d) Find a solution of the initial-value problem

$$y' = -y^2 \quad y(0) = 0.5$$

6. (a) What can you say about the graph of a solution of the equation  $y' = xy^3$  when  $x$  is close to 0? What if  $x$  is large?  
(b) Verify that all members of the family  $y = (c - x^2)^{-1/2}$  are solutions of the differential equation  $y' = xy^3$ .  
(c) Graph several members of the family of solutions on a common screen. Do the graphs confirm what you predicted in part (a)?  
(d) Find a solution of the initial-value problem

$$y' = xy^3 \quad y(0) = 2$$

7. **Logistic growth** A population is modeled by the differential equation

$$\frac{dN}{dt} = 1.2N \left( 1 - \frac{N}{4200} \right)$$

where  $N(t)$  is the number of individuals at time  $t$  (measured in days).

- (a) For what values of  $N$  is the population increasing?
- (b) For what values of  $N$  is the population decreasing?
- (c) What are the equilibrium solutions?

8. **The Fitzhugh-Nagumo model** for the electrical impulse in a neuron states that, in the absence of relaxation effects, the electrical potential in a neuron  $v(t)$  obeys the differential

equation

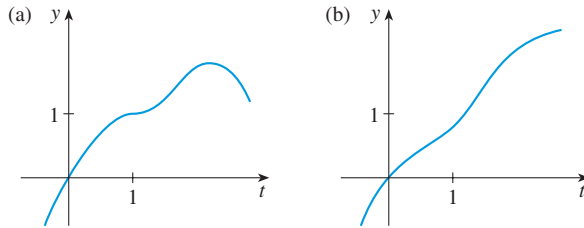
$$\frac{dv}{dt} = -v[v^2 - (1 + a)v + a]$$

where  $a$  is a constant and  $0 < a < 1$ .

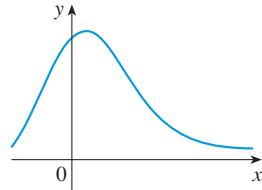
- For what values of  $v$  is  $v$  unchanging (that is,  $dv/dt = 0$ )?
- For what values of  $v$  is  $v$  increasing?
- For what values of  $v$  is  $v$  decreasing?

9. Explain why the functions with the given graphs *can't* be solutions of the differential equation

$$\frac{dy}{dt} = e^t(y - 1)^2$$



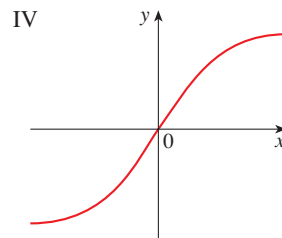
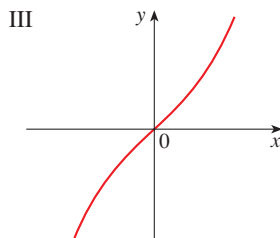
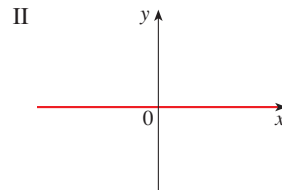
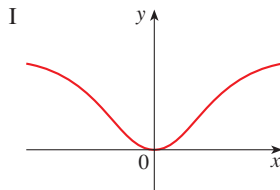
10. The function with the given graph is a solution of one of the following differential equations. Decide which is the correct equation and justify your answer.



- A.  $y' = 1 + xy$     B.  $y' = -2xy$     C.  $y' = 1 - 2xy$

11. Match the differential equations with the solution graphs labeled I–IV. Give reasons for your choices.

- $y' = 1 + x^2 + y^2$
- $y' = xe^{-x^2 - y^2}$
- $y' = \frac{1}{1 + e^{x^2 + y^2}}$
- $y' = \sin(xy) \cos(xy)$



12. **Von Bertalanffy's equation** states that the rate of growth in length of an individual fish is proportional to the difference between the current length  $L$  and the asymptotic length  $L_\infty$  (in cm).

- Write a differential equation that expresses this idea.
- Make a rough sketch of the graph of a solution to a typical initial-value problem for this differential equation.

**13–15 Drug dissolution** Differential equations have been used extensively in the study of drug dissolution for patients given oral medications. The three simplest equations used are the zero-order kinetic equation, the Noyes-Whitney equation, and the Weibull equation. All assume that the initial concentration is zero but make different assumptions about how the concentration increases over time during the dissolution of the medication.

13. The **zero-order kinetic equation** states that the rate of change in the concentration of drug  $c$  (in mg/mL) during dissolution is governed by the differential equation

$$\frac{dc}{dt} = k$$

where  $k$  is a positive constant. Is this differential equation pure-time, autonomous, or nonautonomous? State in words what this differential equation says about how drug dissolution occurs. What is the solution of this differential equation with the initial condition  $c(0) = 0$ ?

14. The **Noyes-Whitney equation** for the dynamics of the drug concentration is

$$\frac{dc}{dt} = k(c_s - c)$$

where  $k$  and  $c_s$  are positive constants. Is this differential equation pure-time, autonomous, or nonautonomous? State in words what this differential equation says about how drug dissolution occurs. Verify that  $c = c_s(1 - e^{-kt})$  is the solution to this equation for the initial condition  $c(0) = 0$ .

15. The **Weibull equation** for the dynamics of the drug concentration is

$$\frac{dc}{dt} = \frac{k}{t^b} (c_s - c)$$

where  $k$ ,  $c_s$ , and  $b$  are positive constants and  $b < 1$ . Notice that this differential equation is undefined when  $t = 0$ . Is this differential equation pure-time, autonomous, or nonautonomous? State in words what this differential equation says about how drug dissolution occurs. Verify that

$$c = c_s(1 - e^{-\alpha t^{1-b}})$$

is a solution for  $t \neq 0$ , where  $\alpha = k/(1 - b)$ .

16. **The logistic differential equation** Suppose that the per capita growth rate of a population of size  $N$  declines linearly from a value of  $r$  when  $N = 0$  to a value of 0 when  $N = K$ .

Show that the differential equation for  $N$  is

$$\frac{dN}{dt} = r \left( 1 - \frac{N}{K} \right) N$$

- 17. Modeling yeast populations** Use the fact that the per capita growth rate of the yeast population in Table 1 is  $0.55 - 0.0026N$  to show that, in terms of the logistic equation (4),  $r = 0.55$  and  $K \approx 210$ .

- 18. Modeling yeast populations (cont.)** Verify that

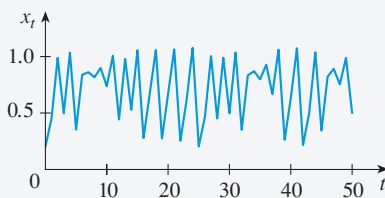
$$N(t) = \frac{42e^{0.55t}}{209.8 + 0.2e^{0.55t}}$$

is an approximate solution of the differential equation

$$\frac{dN}{dt} = (0.55 - 0.0026N)N$$

## PROJECT Chaotic Blowflies and the Dynamics of Populations

BB



**FIGURE 1**

$x$  is plotted against time with  $x_0 = 0.1$  and  $R_{\max} = 3.89$ .

In Section 1.6 we explored the dynamics of the logistic difference equation. After some simplification, the population size in successive times steps was given by the recursion

$$(1) \quad x_{t+1} = R_{\max} x_t (1 - x_t)$$

where  $R_{\max}$  is a positive constant. See Equation 1.6.7. For large enough values of  $R_{\max}$  the recursion exhibits very complicated behavior, as shown in Figure 1. In fact, Equation 1 is famous for being one of the simplest recursions that exhibits chaotic dynamics.<sup>1</sup>

The plots for the logistic differential equation that we have seen in Section 7.1 do not exhibit this type of complicated behavior. Here we explore why. To do so, we will derive the logistic differential equation from the logistic difference equation.

1. In Section 1.6 we obtained Equation 1 by starting with the equation

$$N_{t+1} = [1 + r(1 - N_t/K)] N_t$$

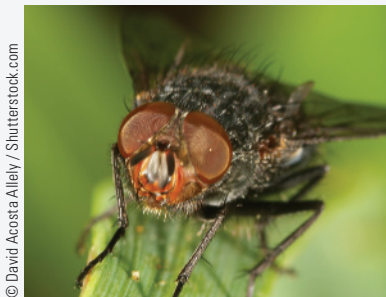
(See Equation 1.6.5.) If the time interval is of length  $h$  instead, where  $h < 1$ , then this equation becomes

$$N_{t+h} = [1 + rh(1 - N_t/K)] N_t$$

Use this result to derive a differential equation for  $N$  by writing an expression for  $\frac{N_{t+h} - N_t}{h}$  and then letting  $h \rightarrow 0$ .

2. Show that with the change of variables  $y = N/K$  the differential equation from Problem 1 can be written as  $dy/dt = ry(1 - y)$ .
3. In Section 7.4 we will learn how to solve differential equations like the one in Problem 2. If  $y(0) = y_0$ , the solution is  $y(t) = y_0/[e^{-rt} + y_0(1 - e^{-rt})]$ . Sketch this solution for different choices of  $y_0$  and  $r$ . This solution can never exhibit the sort of behavior of Equation 1 that is displayed in Figure 1. Explain why from a biological standpoint.

The reason for the complicated dynamics in Figure 1 is the existence of a time-lag between the current population size  $x_t$  and its effects on population regulation. This allows the population to overshoot its carrying capacity. Once an overshoot occurs, a dramatic population decline will ensue. The resulting low population size then sets the stage for a large population rebound and another overshoot of the carrying capacity. Some

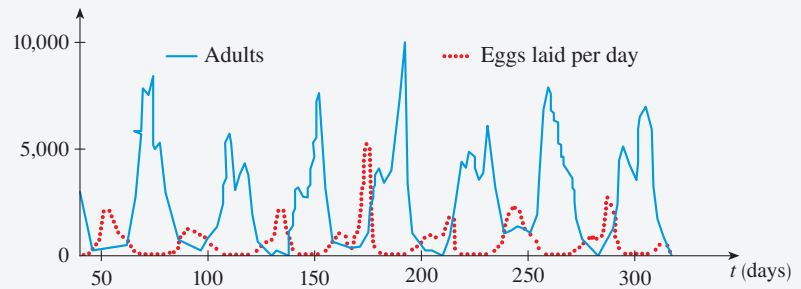


A blowfly

1. R. May, "Simple Mathematical Models with Very Complicated Dynamics," *Nature* 261 (1976): 459–67.

insect species, like blowflies, are believed to be subject to such dynamics, as shown in Figure 2. The exploration of chaos in nature is a fascinating area of research.<sup>2</sup>

**FIGURE 2**  
Irregular fluctuations in a lab  
population of blowflies  
Source: Adapted from W. Gurney et al.,  
“Nicholson’s Blowflies Revisited,”  
*Nature* 287 (1980): 17–21.



2. A. Hastings et al., “Chaos in Ecology: Is Mother Nature a Strange Attractor?” *Annual Review of Ecology, Evolution, and Systematics* 24 (1993): 1–33.

## 7.2 Phase Plots, Equilibria, and Stability

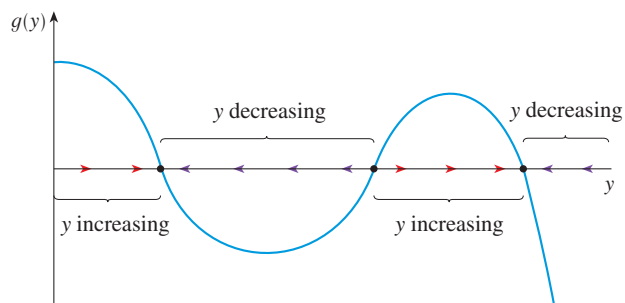
We now examine some techniques for analyzing the behavior of differential equations. Here we focus on a graphical technique known as a *phase plot*.

### ■ Phase Plots

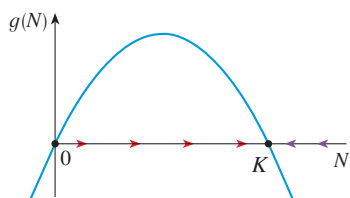
**Phase plots** provide a way to visualize the dynamics of autonomous differential equations, to locate their equilibria, and to determine the stability properties of these equilibria. Consider the autonomous differential equation

$$\frac{dy}{dt} = g(y)$$

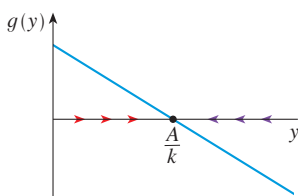
To construct a phase plot we graph the right side of the differential equation,  $g(y)$ , as a function of the dependent variable  $y$ . Where this plot lies above the horizontal axis,  $y'(t) > 0$  and so  $y$  is increasing. Where it lies below the horizontal axis,  $y'(t) < 0$  and so  $y$  is decreasing. Points where the plot crosses the axis correspond to values of the variable at which  $y'(t) = 0$ . We can use these considerations to place arrows on the horizontal axis indicating the direction of change in  $y$ , as shown in Figure 1.



**FIGURE 1**  
A typical phase plot



**FIGURE 2**  
Phase plot for logistic model



**FIGURE 3**  
Phase plot for Example 2

**EXAMPLE 1** | The logistic equation Construct a phase plot for the logistic growth model  $dN/dt = r(1 - N/K)N$  assuming that  $r > 0$ .

**SOLUTION** We need to plot  $g(N) = r(1 - N/K)N$  as a function of  $N$ . This is a parabola opening downward, crossing the horizontal axis at  $N = 0$  and  $N = K$ . In Figure 2 we can see that  $N$  increases when taking on values between 0 and  $K$  and decreases when taking on values greater than  $K$ .

**EXAMPLE 2** | Modeling intravenous drug delivery In Example 7.1.2 we developed a model for the dynamics of the concentration of a drug in the bloodstream. If  $A$  is a positive constant representing the rate of drug delivery through an intravenous supply and  $k$  is a positive constant related to the rate of metabolism of the drug, we obtained

$$\frac{dy}{dt} = A - ky$$

where  $y(t)$  is the concentration of drug in the bloodstream at time  $t$  (in mg/mL). Construct a phase plot for this model.

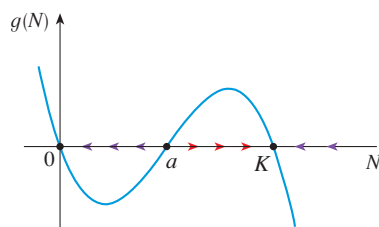
**SOLUTION** We plot  $g(y) = A - ky$  as a function of  $y$ . This is a straight line with slope  $-k$ , intersecting the horizontal axis at  $y = A/k$ . We see that  $y$  increases when it is less than  $A/k$  and decreases when greater than  $A/k$ . (See Figure 3.) This agrees with the plot of  $y$  as a function of time in Figure 7.1.9.

**EXAMPLE 3** | **BB** The Allee effect Some populations decline to extinction once their size is less than a critical value. For example, if the population size is too small, then individuals might have difficulty finding mates for reproduction. This is referred to as an *Allee effect* after the American ecologist Warder Clyde Allee (1885–1955). A simple extension of the logistic model that incorporates this effect is given by

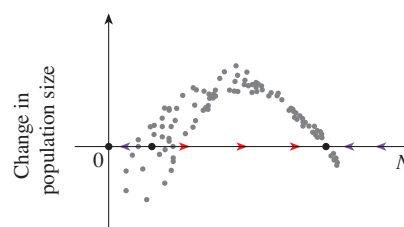
$$\frac{dN}{dt} = r(N - a)\left(1 - \frac{N}{K}\right)N$$

where  $0 < a < K$ . Construct a phase plot assuming that  $r > 0$ .

**SOLUTION** We plot  $g(N) = r(N - a)(1 - N/K)N$  as a function of  $N$ . This is a cubic polynomial whose graph crosses the horizontal axis at  $N = 0$ ,  $N = a$ , and  $N = K$ . The graph lies below the horizontal axis for values of  $N$  between 0 and  $a$  and for values of  $N > K$ . [See Figure 4(a).] Therefore  $N$  will approach 0 if it starts between 0 and  $a$ , whereas it will approach  $K$  if it starts anywhere greater than  $a$ . Figure 4(b) displays data for the phase plot of an experimental microbial population.



(a) Phase plot for an Allee effect



(b) Data for the phase plot of a microbial population

Source: Part (b) adapted from L. Dai et al.,  
“Generic Indicators for Loss of Resilience before  
a Tipping Point Leading to Population Collapse,”  
*Science* 336 (2012): 1175–77.

### ■ Equilibria and Stability

Phase plots provide information about how the dependent variable changes, as well as values of the variable at which no change occurs. Such values are referred to as *equilibria*.

**Definition** Consider the autonomous differential equation

$$(1) \quad \frac{dy}{dt} = g(y)$$

An **equilibrium** solution is a constant value of  $y$  (denoted  $\hat{y}$ ) such that  $dy/dt = 0$  when  $y = \hat{y}$ .

Equilibria are found by determining values of  $\hat{y}$  that satisfy  $g(\hat{y}) = 0$ . They correspond to places where the phase plot crosses the horizontal axis.

**EXAMPLE 4** | The logistic equation (continued) Show that  $\hat{N} = 0$  and  $\hat{N} = K$  are equilibria of the logistic growth model from Example 1.

**SOLUTION** Substituting  $\hat{N} = 0$  and  $\hat{N} = K$  into the equation  $g(N) = r(1 - N/K)N$  gives  $g(\hat{N}) = 0$  in both cases. In the yeast data of Section 7.1,  $\hat{N} = K$  corresponds to the steady number of yeast cells reached as the experiment progressed. ( $\hat{N} = 0$  corresponds to the absence of yeast.) ■

**EXAMPLE 5** | Modeling intravenous drug delivery (continued) Find all equilibria of the model in Example 2.

**SOLUTION** We seek values of  $\hat{y}$  that make  $g(\hat{y}) = 0$ ; that is, values of  $\hat{y}$  that satisfy the equation  $A - k\hat{y} = 0$ . The only solution is  $\hat{y} = A/k$ . Again this agrees with the plot of  $y$  as a function of time  $t$  in Figure 7.1.9. ■

In addition to providing a way to visualize equilibria, phase plots provide information about their stability properties.

**Definition** An equilibrium  $\hat{y}$  of differential equation (1) is **locally stable** if  $y$  approaches the value  $\hat{y}$  as  $t \rightarrow \infty$  for all initial values of  $y$  sufficiently close to  $\hat{y}$ .

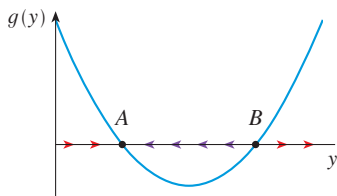
An equilibrium that is not stable is referred to as **unstable**. In Example 1, the arrows on the horizontal axis of Figure 2 reveal that  $\hat{N} = 0$  is an unstable equilibrium whereas  $\hat{N} = K$  is a locally stable equilibrium. In Example 2, the arrows in Figure 3 reveal that  $\hat{y} = A/k$  is a locally stable equilibrium.

**EXAMPLE 6** | **BB** The Allee effect (continued) Find all equilibria for the model of an Allee effect in Example 3 and determine their stability properties from the phase plot in Figure 4(a).

**SOLUTION** We need to find the values of  $\hat{N}$  that satisfy the equation

$$r(\hat{N} - a)(1 - \hat{N}/K)\hat{N} = 0$$

We can see that these are  $\hat{N} = 0$ ,  $\hat{N} = a$ , and  $\hat{N} = K$ . These are the points at which the phase plot in Figure 4(a) crosses the horizontal axis. From the arrows on the figure we can also see that both  $\hat{N} = 0$  and  $\hat{N} = K$  are locally stable whereas  $\hat{N} = a$  is unstable—no matter how close we start  $N$  to the value  $a$ , it always moves farther away from  $a$  as time passes.



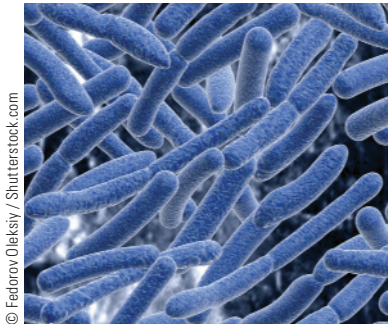
**FIGURE 5**  
A phase plot

We have used phase plots to determine stability graphically, but these plots also provide a guide for using calculus to do so. Let's examine the phase plot in Figure 5. What property of the plot guarantees that the equilibrium  $A$  is locally stable? This equilibrium is locally stable because the plot lies above the horizontal axis for values of  $y$  less than  $A$  and it lies below the horizontal axis for values of  $y$  greater than  $A$  but close to  $A$ . Specifically, the plot crosses the horizontal axis from above to below as it passes through  $A$ . This will be true if the slope of the plot at  $A$  is negative. Similarly, convince yourself that  $B$  is unstable because the slope of the plot is positive at this equilibrium.

**Local Stability Criterion** Suppose that  $\hat{y}$  is an equilibrium of the differential equation

$$\frac{dy}{dt} = g(y)$$

Then  $\hat{y}$  is *locally stable* if  $g'(\hat{y}) < 0$ , and  $\hat{y}$  is *unstable* if  $g'(\hat{y}) > 0$ . If  $g'(\hat{y}) = 0$ , then the analysis is inconclusive.



**FIGURE 6**  
Cross-feeding bacteria

**EXAMPLE 7 | Population genetics** Two bacterial strains sometimes feed on chemicals excreted by one another: strain A feeds on chemicals produced by strain B, and vice versa. This phenomenon is referred to as *cross-feeding*. Suppose that two strains of bacteria are engaged in cross-feeding (strain 1 and strain 2). Exercise 17 asks you to show that, for a relatively simple model of cross-feeding, the frequency  $p(t)$  of the strain 1 bacteria is governed by the differential equation

$$\frac{dp}{dt} = p(1 - p)[\alpha(1 - p) - \beta p]$$

where  $\alpha$  and  $\beta$  are positive constants. Suppose that  $\alpha = 1$  and  $\beta = 2$ . Then the differential equation simplifies to

$$\frac{dp}{dt} = p(1 - p)[(1 - p) - 2p] = p(1 - p)(1 - 3p)$$

- Find all equilibria.
- Determine the stability properties of each equilibrium found in part (a).

### SOLUTION

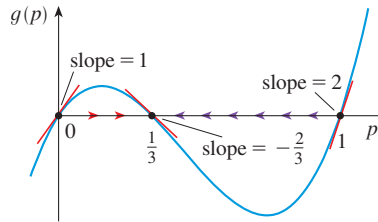
- Equilibria are values of  $\hat{p}$  satisfying the equation

$$\hat{p}(1 - \hat{p})(1 - 3\hat{p}) = 0$$

This gives  $\hat{p} = 0$ ,  $\hat{p} = 1$ , and  $\hat{p} = \frac{1}{3}$ .

- We first need to calculate the derivative of  $g(p) = p(1 - p)(1 - 3p)$  with respect to  $p$ . After some simplification we obtain

$$g'(p) = 1 - 8p + 9p^2$$



**FIGURE 7**  
Phase plot for cross-feeding model

We then need to evaluate this at each of the equilibria.

- $\hat{p} = 0$ :  $g'(0) = 1 - 8(0) + 9(0)^2 = 1$   
which is positive, meaning that  $\hat{p} = 0$  is unstable.
- $\hat{p} = 1$ :  $g'(1) = 1 - 8(1) + 9(1)^2 = 2$   
which is positive, meaning that  $\hat{p} = 1$  is unstable.
- $\hat{p} = \frac{1}{3}$ :  $g'(\frac{1}{3}) = 1 - 8(\frac{1}{3}) + 9(\frac{1}{3})^2 = -\frac{2}{3}$   
which is negative, meaning that  $\hat{p} = \frac{1}{3}$  is locally stable.

These results suggest that, over time, we expect the frequency of strain 1 bacteria in the population to approach  $\hat{p} = \frac{1}{3}$  as indicated by the arrows in Figure 7. ■

### ■ A Mathematical Derivation of the Local Stability Criterion

We obtained the local stability criterion from graphical considerations, but we can also derive it more rigorously. Consider the autonomous differential equation

$$\frac{dy}{dt} = g(y)$$

Suppose that  $\hat{y}$  is an equilibrium [that is,  $g(\hat{y}) = 0$ ]. To determine if this equilibrium is locally stable, we need to determine whether  $y$  will approach the value  $\hat{y}$  provided that we start the variable sufficiently close to this value.

Let's consider starting the value of  $y$  a small distance  $\varepsilon$  from  $\hat{y}$ . In other words,  $\varepsilon(t) = y(t) - \hat{y}$ . If  $\hat{y}$  is locally stable, then the magnitude of  $\varepsilon(t)$  must decrease as time passes.

We can derive a differential equation that governs the dynamics of  $\varepsilon(t)$  by differentiating  $\varepsilon(t)$  with respect to  $t$  and using the differential equation for  $y$ :

$$\frac{d\varepsilon}{dt} = \frac{d}{dt}[y(t) - \hat{y}] = \frac{dy}{dt} = g(y) = g(\varepsilon + \hat{y})$$

where we have obtained the final equality by using the fact that  $\varepsilon(t) = y(t) - \hat{y}$ . Now, provided that we start the value of  $y$  near  $\hat{y}$ ,  $\varepsilon$  will be small and we can therefore approximate the function  $g(\varepsilon + \hat{y})$  using a linear approximation near the value  $\varepsilon = 0$  (see Section 3.8). In other words, we can write

$$g(\varepsilon + \hat{y}) \approx g(\hat{y}) + g'(\hat{y}) \cdot \varepsilon$$

provided that  $\varepsilon$  is small. Furthermore, because  $\hat{y}$  is an equilibrium we know that  $g(\hat{y}) = 0$ , and therefore the differential equation for  $\varepsilon$  simplifies to

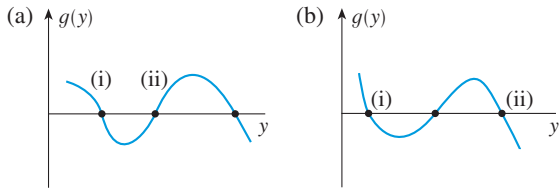
$$\frac{d\varepsilon}{dt} \approx g'(\hat{y}) \cdot \varepsilon$$

At this point we note that the quantity  $g'(\hat{y})$  is a constant (that is, it is not time-varying). As a result,  $\varepsilon$  approximately obeys a differential equation of the form  $d\varepsilon/dt = k\varepsilon$ , where  $k = g'(\hat{y})$  is a constant. This suggests that  $\varepsilon(t)$  grows over time (meaning equilibrium  $\hat{y}$  is unstable) if  $g'(\hat{y}) > 0$ . And  $\varepsilon(t)$  decays over time (meaning equilibrium  $\hat{y}$  is locally stable) if  $g'(\hat{y}) < 0$ .

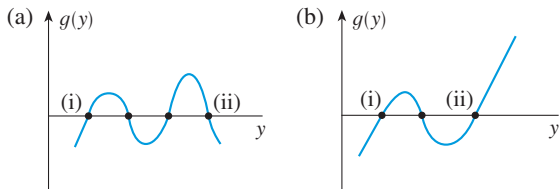
## EXERCISES 7.2

1–3 Consider the differential equation  $dy/dt = g(y)$ .

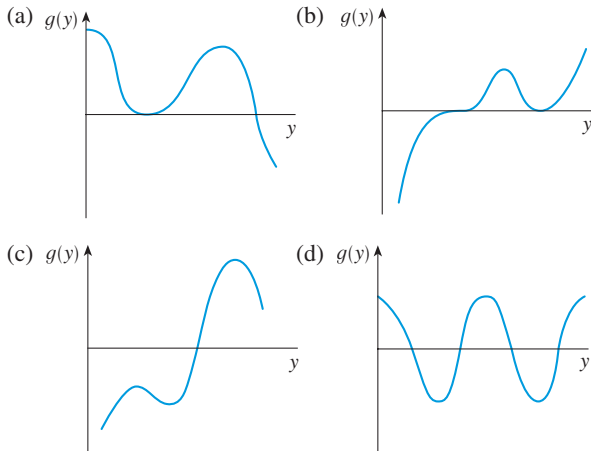
1. For each graph, determine whether the equilibria (i) and (ii) are locally stable.



2. For each graph, determine whether the equilibria (i) and (ii) are unstable.



3. Complete the phase plot for each graph by locating the equilibria and indicating the direction in which  $y$  changes on the horizontal axis.



4–6 Find all equilibria of the autonomous differential equation and construct the phase plot.

4. (a)  $y' = y^2 - 2$   
 (b)  $y' = \frac{y-3}{y+9}, \quad y \geq 0$   
 (c)  $y' = y(3-y)(25-y^2)$
5. (a)  $y' = y + 2 \ln y, \quad y > 0$   
 (b)  $y' = y^3 - a, \quad a \geq 0$   
 (c)  $y' = \frac{5}{2+y}, \quad y \geq 0$

6. (a)  $y' = \frac{y^2 - a}{y + 1}, \quad 0 \leq a < 1, y > -1$   
 (b)  $y' = y(a-y)^2(b-y), \quad b > a \geq 0$   
 (c)  $y' = \frac{a}{b+y} - 1, \quad a > b > 0, y \geq 0$

7–8 Find all equilibria and use the local stability criterion to determine if each is locally stable or unstable. Then construct the phase plot.

7. (a)  $y' = 5 - 3y$  (b)  $y' = 2y - 3y^2$
8. (a)  $y' = 5y(2e^{-y} - 1)$   
 (b)  $y' = y^{5/3} - 2y$  (assume  $y \geq 0$ )

9. Find the equilibria for the differential equation and determine the values of  $a$  for which each equilibrium is locally stable. Assume  $a \neq 0$ .

- (a)  $y' = 1 + ay$   
 (b)  $y' = 1 - e^{-ay}$   
 (c)  $y' = ae^y \cos y, \quad 0 < y < \pi$   
 (d)  $y' = y(a-y)$

10. **The Allee effect** For the model of population dynamics from Example 3, use the local stability criterion to verify that  $\hat{N} = 0$  and  $\hat{N} = K$  are locally stable whereas  $\hat{N} = a$  is unstable.

11. Suppose that the population dynamics of a species obeys a modified version of the logistic differential equation having the following form:

$$\frac{dN}{dt} = r \left( 1 - \frac{N}{K} \right)^2 N$$

where  $r \neq 0$  and  $K > 0$ .

- (a) Show that  $\hat{N} = 0$  and  $\hat{N} = K$  are equilibria.  
 (b) For which values of  $r$  is the equilibrium  $\hat{N} = 0$  unstable?  
 (c) Apply the local stability criterion to the equilibrium  $\hat{N} = K$ . What do you think your answer means about the stability of this equilibrium? (Note: This is an example in which the local stability criterion is inconclusive.)  
 (d) Construct two phase plots, one for the case where  $r > 0$  and the other for  $r < 0$ , and determine the stability of  $\hat{N} = K$  in each case. Does the answer match your reasoning in part (c)?

12. **Infectious disease dynamics** The spread of an infectious disease, such as influenza, is often modeled using the following autonomous differential equation:

$$\frac{dI}{dt} = \beta I(N - I) - \mu I$$

where  $I$  is the number of infected people,  $N$  is the total size of the population being modeled,  $\beta$  is a constant determin-

ing the rate of transmission, and  $\mu$  is the rate at which people recover from infection.

- Suppose  $\beta = 0.01$ ,  $N = 1000$ , and  $\mu = 2$ . Find all equilibria.
- For the equilibria in part (a), determine whether each is stable or unstable.
- Leaving the constants unspecified, what are the equilibria of the model in terms of these constants?
- The epidemiological quantity  $R_0$  that was introduced in Example 1.5.5 is calculated by rearranging the condition for  $\hat{I} = 0$  to be unstable, giving an inequality of the form  $R_0 > 1$ . Show that  $R_0$  can be written as  $R_0 = \beta N / \mu$ .

- 13. Harvesting of renewable resources** Suppose a population grows according to the logistic equation but is subject to a constant per capita harvest rate of  $h$ . If  $N(t)$  is the population size at time  $t$ , the population dynamics are

$$\frac{dN}{dt} = r \left( 1 - \frac{N}{K} \right) N - hN$$

Different values of  $h$  result in different equilibrium population sizes; if  $h$  is large enough, we might expect extinction.

- Suppose  $r = 2$  and  $K = 1000$ . Find all equilibria. [Hint: One will be a function of  $h$ .]
- For the nonzero equilibrium in part (a), what is the critical value of  $h$  greater than which the population will go extinct?
- Determine the values of  $h$  that make the nonzero equilibrium in part (a) locally stable. Assume  $h \neq 2$ .

- 14. Harvesting of renewable resources** Suppose a population grows according to the logistic equation but is subject to a constant total harvest rate of  $H$ . If  $N(t)$  is the population size at time  $t$ , the population dynamics are

$$\frac{dN}{dt} = r \left( 1 - \frac{N}{K} \right) N - H$$

Different values of  $H$  result in different equilibrium population sizes; if  $H$  is large enough, we might expect extinction.

- Suppose  $r = 2$ ,  $K = 1000$ , and  $H = 100$ . Find all equilibria.
- Determine whether each of the equilibria in part (a) is locally stable or unstable. Is the population predicted to go extinct?

- 15. Levins' metapopulation model** Many species are made up of several small subpopulations that occasionally go extinct but that are subsequently recolonized. The entire collection of subpopulations is referred to as a metapopulation. One way to model this phenomenon is to keep track of only the fraction of subpopulations that are currently not extinct. Suppose  $p(t)$  is the fraction of subpopulations that are not extinct at time  $t$ . The Levins model states that  $p(t)$

obeys the differential equation

$$\frac{dp}{dt} = cp(1 - p) - mp$$

where  $c$  and  $m$  are positive constants reflecting the colonization and extinction rates, respectively. (See also Review Exercise 15 on page 482.) Assume  $c \neq m$ .

- What are the equilibria of this model in terms of the constants?
- What is the condition on the constants for the nonzero equilibrium in part (a) to lie between zero and one? Interpret this condition.
- What are the conditions on the constants for the nonzero equilibrium in part (a) to be locally stable? Assume  $m \neq c$ .

- 16. Bacteria population genetics** Suppose there are two bacterial strains 1 and 2, each undergoing growth according to the differential equations

$$\frac{dN_1}{dt} = r_1 N_1 \quad \text{and} \quad \frac{dN_2}{dt} = r_2 N_2$$

respectively, where  $r_1 \neq r_2$ . Define

$$p(t) = \frac{N_1(t)}{N_1(t) + N_2(t)}$$

to be the frequency of strain 1 at time  $t$ .

- Differentiate  $p$  using its definition to show that  $p$  obeys the differential equation

$$\frac{dp}{dt} = sp(1 - p)$$

where  $s = r_1 - r_2$ .

- What are the equilibria of the differential equation of part (a)? For what values of  $s$  is each locally stable?

- 17. Bacterial cross-feeding** The differential equation from Exercise 16 can be extended to model the effects of bacterial cross-feeding. Suppose that the growth rate of strain 1 ( $r_1$ ) is zero when the frequency of strain 2 is zero and that it increases linearly to a maximum value of  $\alpha$  when the frequency of strain 2 is 1. Likewise, suppose that the growth rate of strain 2 ( $r_2$ ) is zero when the frequency of strain 1 is zero and that it increases linearly to a maximum value of  $\beta$  when the frequency of strain 1 is one. Show that these assumptions, combined with the differential equation from Exercise 16 (a), result in the cross-feeding model of Example 7.

- 18. Bacterial cross-feeding (cont.)** The differential equation derived in Exercise 17 (and from Example 7) is

$$\frac{dp}{dt} = p(1 - p)[\alpha(1 - p) - \beta p]$$

where  $\alpha$  and  $\beta$  are both positive constants.

- Find all equilibria.
- Determine the values of  $\alpha$  and  $\beta$  for which each equilibrium in part (a) is locally stable.

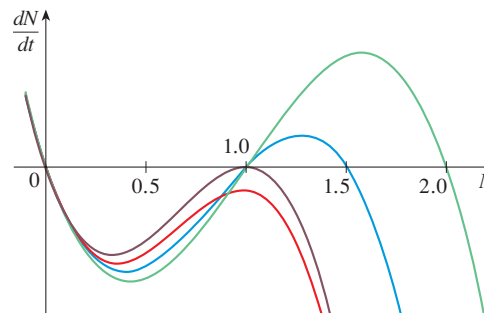
**19. Mutation-selection balance** The population-genetic differential equation in Exercise 16 assumes there is no mutation. Suppose that strain 1 bacteria mutate to strain 2 at a per capita rate of  $\mu$ , but otherwise their dynamics are exactly as given in Exercise 16.

- What is the resulting differential equation for  $p$ ?
- Determine the equilibria of the differential equation from part (a).
- Determine the constant values under which each equilibrium in part (b) is locally stable. Assume  $s \neq \mu$ .

**20. Catastrophic population collapse** The graph depicting the rate of change of a population as a function of population size is given by the green curve in the figure, with  $N$  measured in thousands of individuals. Suppose the population size starts between 1000 and 2000 individuals.

- What is the predicted population size in the long term?
- Now suppose that habitat degradation begins and that  $K$  is a constant quantifying habitat quality (sometimes referred to as the *carrying capacity*).  $K = 2$  represents

a pristine habitat and corresponds to the green curve. As  $K$  decreases, the curve changes continuously, from green to blue to purple to red. Purple corresponds to  $K = 1$ . Sketch a curve of the long-term population size as a function of  $K$ , depicting the main qualitative features. You can assume that the curve is linear except at points of discontinuity.



## PROJECT Catastrophic Population Collapse: An Introduction to Bifurcation Theory

BB



Overexploitation of fish stocks by commercial fisheries can cause catastrophic population collapse.

Biological populations exhibit a wide range of dynamics, from stable population sizes, to oscillatory dynamics, to extinction. As conditions change—for example, through global warming, pollution, or increased harvesting pressure—populations can switch from one of these dynamical regimes to another. When this occurs, a *bifurcation* is said to have happened. **Bifurcation theory** provides a set of tools from the study of differential equations that we can use to study such phenomena.

Consider the following model for the dynamics of a population of size  $N$  (measured as number of individuals  $\times 10^4$ ) over time (in months) that is subject to harvesting:

$$(1) \quad \frac{dN}{dt} = \left(1 - \frac{N}{2}\right)N - hN$$

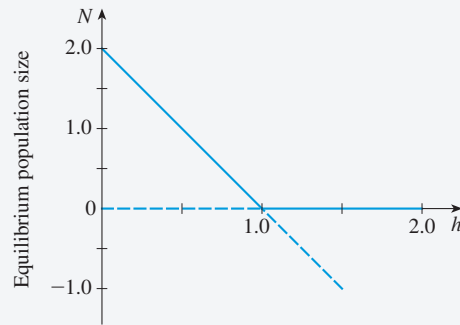
The population grows according to a logistic equation in the absence of harvesting and  $h$  is a constant per capita harvest rate (see also Exercise 7.2.13).

- Find all equilibria and determine the values of  $h$  for which each is stable or unstable.

A **bifurcation plot** is a plot of all equilibria of a differential equation as a function of a constant of interest (called the bifurcation parameter). Solid curves are used to represent locally stable equilibria and dashed curves represent unstable equilibria.

- Plot the equilibria from Problem 1 as a function of  $h$  for  $0 \leq h \leq 2$ , using solid and dashed curves as described. In other words, construct the bifurcation plot.

You should obtain the plot in Figure 1.



**FIGURE 1**

Bifurcation plot for Equation 1

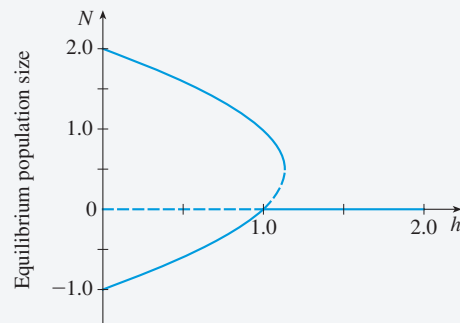
- Describe in words what happens to the predicted population size as  $h$  increases continuously from  $h = 0$  to  $h = 2$ . (You can assume that the initial population size is positive.) Equation 1 exhibits what is called a *transcritical bifurcation*. As the bifurcation parameter is increased, two equilibria move continuously, eventually merging and exchanging stability properties before once again diverging.

Now consider the following alternative model:

$$(2) \quad \frac{dN}{dt} = \left(1 - \frac{N}{2}\right)N - h \frac{N}{1 + N}$$

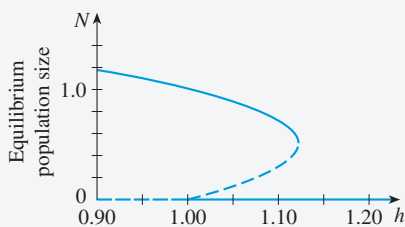
The only difference between (1) and (2) is in the form of the loss rate through harvesting.

- Describe, in words, the form of the loss rate through harvesting as a function of population size in Equation 2. Provide an explanation for why this form is a reasonable way to model harvesting.
- Verify that the bifurcation plot for Equation 2 is as given in Figure 2.



**FIGURE 2**

Bifurcation plot for Equation 2



**FIGURE 3**

Bifurcation plot for Equation 2

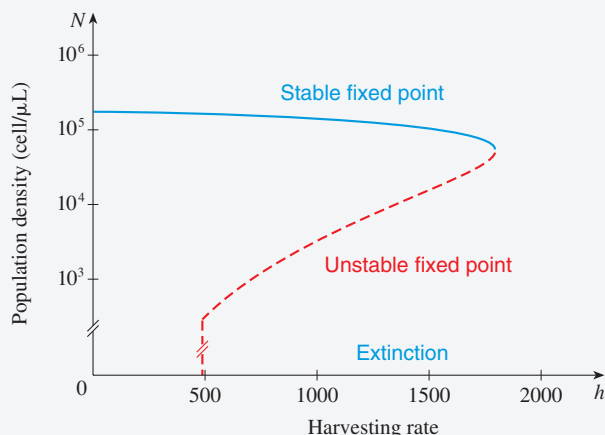
- Describe in words what happens to the predicted population size as  $h$  increases continuously from  $h = 0$  to  $h = 2$ . (You can assume that the initial population size is positive.)

Equation 2 exhibits a *saddle-node bifurcation*: As the bifurcation parameter is increased, two equilibria move continuously, eventually merging and annihilating one another. This can result in an abrupt or discontinuous change in model predictions as the bifurcation parameter changes.

Figure 3 zooms in on the bifurcation plot from Figure 2. You can see that, for values of  $h$  between  $h = 1$  and  $h = 1.125$ , there are two locally stable equilibria separated by

an unstable equilibrium. This type of bifurcation has been observed in real populations by experimentally manipulating the loss rate (see Figure 4).<sup>1</sup>

**FIGURE 4**  
An experimentally measured  
bifurcation plot for a  
microbial population



1. Adapted from L. Dai et al., “Generic Indicators for Loss of Resilience before a Tipping Point Leading to Population Collapse,” *Science* 336 (2012): 1175–77.

## 7.3 Direction Fields and Euler’s Method

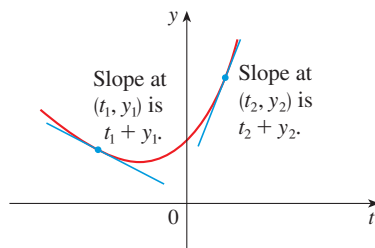
In this section we develop another graphical technique known as a direction field. This will allow us to gain more information about the shapes of solution curves. It will also lead us to Euler’s method for solving differential equations numerically.

### Direction Fields

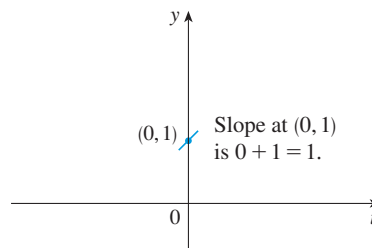
Suppose we are asked to sketch the graph of the solution of the initial-value problem

$$y' = t + y \quad y(0) = 1$$

We don’t know a formula for the solution, so how can we possibly sketch its graph? Let’s think about what the differential equation means. The equation  $y' = t + y$  tells us that the slope at any point  $(t, y)$  on the graph is equal to the sum of the  $t$ - and  $y$ -coordinates of the point (see Figure 1). In particular, because the curve passes through the point  $(0, 1)$ , its slope there must be  $0 + 1 = 1$ . So a small portion of the solution curve near the point  $(0, 1)$  looks like a short line segment through  $(0, 1)$  with slope 1. (See Figure 2.)

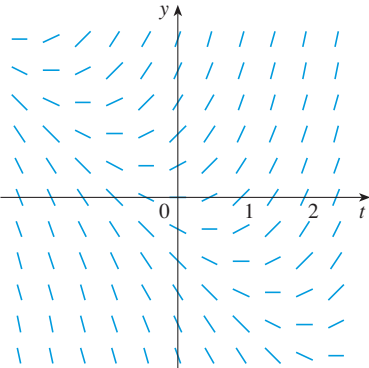


**FIGURE 1**  
A solution of  $y' = t + y$

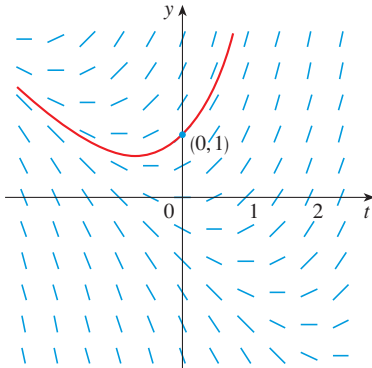


**FIGURE 2**  
Beginning of the solution curve through  $(0, 1)$

As a guide to sketching the rest of the curve, let's draw short line segments at a number of points  $(t, y)$  with slope  $t + y$ . The result, called a *direction field*, is shown in Figure 3. For instance, the line segment at the point  $(1, 2)$  has slope  $1 + 2 = 3$ . The direction field allows us to visualize the general shape of the solution curves by indicating the direction in which the curves proceed at each point.



**FIGURE 3**  
Direction field for  $y' = t + y$



**FIGURE 4**  
The solution curve through  $(0, 1)$

Now we can sketch the solution curve through the point  $(0, 1)$  by following the direction field, as shown in Figure 4. We can visualize  $y(t)$  as being a curve that snakes its way through the direction field, always being parallel to nearby line segments. Such a curve is referred to as a **solution curve**.

In general, suppose we have the first-order differential equation

$$y' = F(t, y)$$

where  $F(t, y)$  is some function in  $t$  and  $y$ . The differential equation says that the slope of a solution curve at a point  $(t, y)$  on the curve is  $F(t, y)$ . If we draw short line segments with slope  $F(t, y)$  at several points  $(t, y)$ , the result is called a **direction field** (or **slope field**). These line segments indicate the direction in which a solution curve is heading.

**EXAMPLE 1**

- (a) Sketch the direction field for the differential equation  $\frac{dy}{dx} = x^2 + y^2 - 1$ .
- (b) Use part (a) to sketch the solution curve that passes through the origin.

**SOLUTION**

- (a) We start by computing the slope at several points, as given in the following table:

$$y' = x^2 + y^2 - 1$$

|     |     |    |    |    |   |   |     |
|-----|-----|----|----|----|---|---|-----|
| $y$ |     |    |    |    |   |   |     |
| 2   | ... | :  | :  | :  | : | : | ... |
| 1   | ... | 4  | 1  | 0  | 1 | 4 | ... |
| 0   | ... | 3  | 0  | -1 | 0 | 3 | ... |
| -1  | ... | 4  | 1  | 0  | 1 | 4 | ... |
| -2  | ... | :  | :  | :  | : | : | ... |
|     | -3  | -2 | -1 | 0  | 1 | 2 | 3   |
|     | $x$ |    |    |    |   |   |     |

Now we draw short line segments with these slopes at the indicated points. In Figure 5 we have used a computer to draw line segments at these and several other points.

**TEC** Module 7.3A shows direction fields and solution curves for a variety of differential equations.

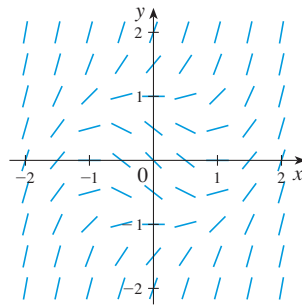


FIGURE 5

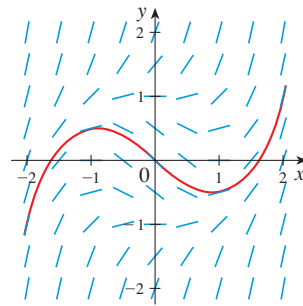


FIGURE 6

(b) We start at the origin and move to the right in the direction of the line segment (which has slope  $-1$ ). We continue to draw the solution curve so that it moves parallel to the nearby line segments. The resulting solution curve is shown in Figure 6. Returning to the origin, we draw the solution curve to the left as well. ■

The more line segments we draw in a direction field, the clearer the solution curves appear. It's tedious to compute slopes and draw line segments for a large number of points by hand, but computers are well suited for this task. Figure 7 shows an even more detailed computer-drawn direction field for the differential equation in Example 1. It enables us to draw, with reasonable accuracy, the solution curves shown in Figure 8 with y-intercepts  $-2$ ,  $-1$ ,  $0$ ,  $1$ , and  $2$ .

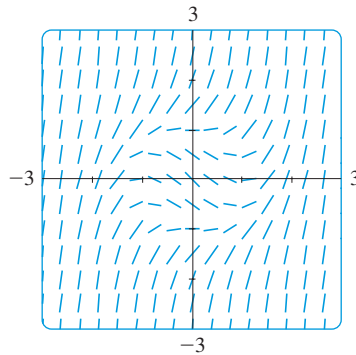


FIGURE 7

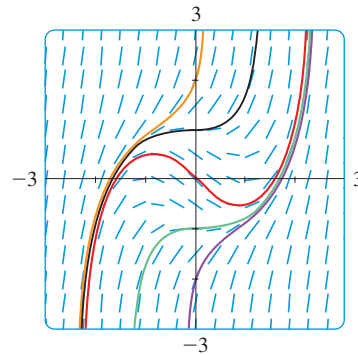


FIGURE 8

**EXAMPLE 2** | Population genetics (continued) The model presented in Example 7.2.7 reduces to

$$\frac{dp}{dt} = p(1 - p)(2 - 5p)$$

when  $\alpha = 2$  and  $\beta = 3$ .

- Draw the direction field for  $0 \leq t \leq 8$  and  $0 \leq p \leq 1$ .
- Identify all equilibria on the plot.
- Sketch the solution curve that starts at  $0.8$  when  $t = 0$ ; that is,  $p(0) = 0.8$ .
- What happens to the solution curve plotted in part (c) as  $t \rightarrow \infty$ ?

### SOLUTION

(a) We start by computing the slope at several points in the following table:

$$p' = p(1 - p)(2 - 5p)$$

| $p$ |       |       |       |       |       |       |       |       |       |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1   | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| 0.8 | -0.32 | -0.32 | -0.32 | -0.32 | -0.32 | -0.32 | -0.32 | -0.32 | -0.32 |
| 0.6 | -0.24 | -0.24 | -0.24 | -0.24 | -0.24 | -0.24 | -0.24 | -0.24 | -0.24 |
| 0.4 | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| 0.2 | 0.16  | 0.16  | 0.16  | 0.16  | 0.16  | 0.16  | 0.16  | 0.16  | 0.16  |
| 0   | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|     | 0     | 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     |
|     | $t$   |       |       |       |       |       |       |       |       |

Notice that all the columns are identical. This is because the independent variable  $t$  does not occur on the right side of the equation for  $p'$  (it is *autonomous*). As a result, the slopes corresponding to two different points with the same  $p$ -coordinate must be equal. Thus, if we know one solution to an autonomous differential equation, then we can obtain infinitely many others just by shifting the graph of the known solution to the right or left, as shown in Figure 9.

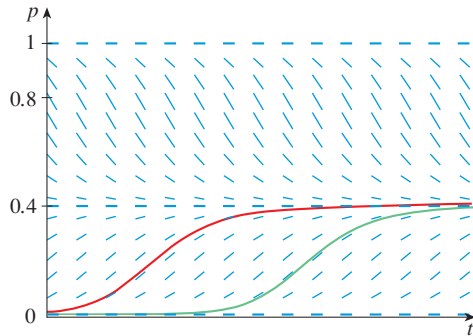


FIGURE 9

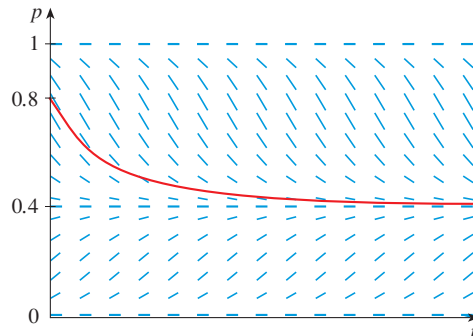


FIGURE 10

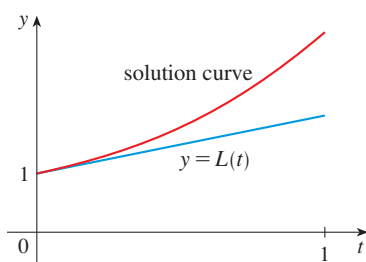
- (b) The equilibria are values of the variable at which the direction field is horizontal. From Figure 10 we can see that equilibria occur at  $\hat{p} = 0$ ,  $\hat{p} = 0.4$ , and  $\hat{p} = 1$ .
- (c) The solution curve corresponding to  $p(0) = 0.8$  is obtained by starting at this point and drawing a curve that moves in the direction of the line segment at this point and then makes its way through the direction field, always remaining parallel to nearby line segments (see Figure 10).
- (d) From the solution curve plotted in part (c) it appears that  $p(t) \rightarrow 0.4$  as  $t \rightarrow \infty$ . ■

### Euler's Method

The basic idea behind direction fields can be used to find numerical approximations to solutions of differential equations. We illustrate the method on the initial-value problem that we used to introduce direction fields:

$$y' = t + y \quad y(0) = 1$$

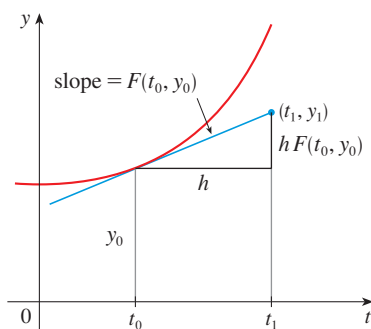
The differential equation tells us that  $y'(0) = 0 + 1 = 1$ , so the solution curve has slope

**FIGURE 11**

First Euler approximation

**Euler**

Leonhard Euler (1707–1783) was the leading mathematician of the mid-18th century and the most prolific mathematician of all time. He was born in Switzerland but spent most of his career at the academies of science supported by Catherine the Great in St. Petersburg and Frederick the Great in Berlin. The collected works of Euler (pronounced *Oiler*) fill about 100 large volumes. As the French physicist Arago said, “Euler calculated without apparent effort, as men breathe or as eagles sustain themselves in the air.” Euler’s calculations and writings were not diminished by raising 13 children or being totally blind for the last 17 years of his life. In fact, when blind, he dictated his discoveries to his helpers from his prodigious memory and imagination. His treatises on calculus and most other mathematical subjects became the standard for mathematics instruction, and the equation  $e^{i\pi} + 1 = 0$  that he discovered brings together the five most famous numbers in all of mathematics.

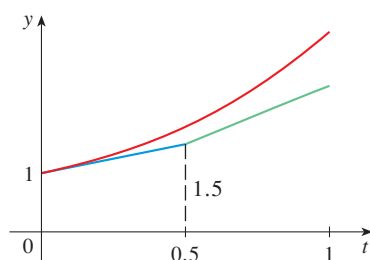
**FIGURE 14**

1 at the point  $(0, 1)$ . As a first approximation to the solution we could use the linear approximation  $L(t) = t + 1$ . In other words, we could use the tangent line at  $(0, 1)$  as a rough approximation to the solution curve (see Figure 11).

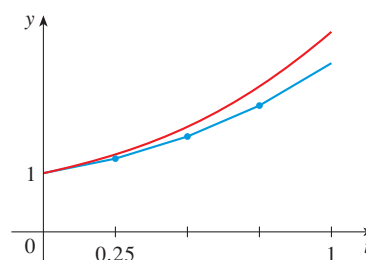
Euler’s idea was to improve on this approximation by proceeding only a short distance along this tangent line and then making a midcourse correction by changing direction as indicated by the direction field. Figure 12 shows what happens if we start out along the tangent line but stop when  $t = 0.5$ . (This horizontal distance traveled is called the *step size*.) Since  $L(0.5) = 1.5$ , we have  $y(0.5) \approx 1.5$  and we take  $(0.5, 1.5)$  as the starting point for a new line segment. The differential equation tells us that  $y'(0.5) = 0.5 + 1.5 = 2$ , so we use the linear function

$$y = 1.5 + 2(t - 0.5) = 2t + 0.5$$

as an approximation to the solution for  $t > 0.5$  (the green segment in Figure 12). If we decrease the step size from 0.5 to 0.25, we get the better Euler approximation shown in Figure 13.

**FIGURE 12**

Euler approximation with step size 0.5

**FIGURE 13**

Euler approximation with step size 0.25

In general, Euler’s method says to start at the point given by the initial value and proceed in the direction indicated by the direction field. Stop after a short time, look at the slope at the new location, and proceed in that direction. Keep stopping and changing direction according to the direction field. Euler’s method does not produce the exact solution to an initial-value problem—it gives approximations. But by decreasing the step size (and therefore increasing the number of midcourse corrections), we obtain successively better approximations to the exact solution. (Compare Figures 11, 12, and 13.)

For the general first-order initial-value problem  $y' = F(t, y)$ ,  $y(t_0) = y_0$ , our aim is to find approximate values for the solution at equally spaced numbers  $t_0, t_1 = t_0 + h, t_2 = t_1 + h, \dots$ , where  $h$  is the step size. The differential equation tells us that the slope at  $(t_0, y_0)$  is  $y' = F(t_0, y_0)$ , so Figure 14 shows that the approximate value of the solution when  $t = t_1$  is

$$y_1 = y_0 + hF(t_0, y_0)$$

Similarly,

$$y_2 = y_1 + hF(t_1, y_1)$$

In general,

$$y_{n+1} = y_n + hF(t_n, y_n)$$

**Euler’s Method** Approximate values for the solution of the initial-value problem  $y' = F(t, y)$ ,  $y(t_0) = y_0$ , with step size  $h$ , at  $t_{n+1} = t_n + h$ , are

$$y_{n+1} = y_n + hF(t_n, y_n) \quad n = 1, 2, 3, \dots$$

Euler's method amounts to approximating the differential equation with a difference equation (see Section 1.6); the smaller the time step, the better the approximation.

**EXAMPLE 3** | Use Euler's method with step size 0.1 to construct a table of approximate values for the solution of the initial-value problem

$$\frac{dy}{dx} = x + y \quad y(0) = 1$$

**SOLUTION** We are given that  $h = 0.1$ ,  $x_0 = 0$ ,  $y_0 = 1$ , and  $F(x, y) = x + y$ . So we have

$$\begin{aligned} y_1 &= y_0 + hF(x_0, y_0) = 1 + 0.1(0 + 1) = 1.1 \\ y_2 &= y_1 + hF(x_1, y_1) = 1.1 + 0.1(0.1 + 1.1) = 1.22 \\ y_3 &= y_2 + hF(x_2, y_2) = 1.22 + 0.1(0.2 + 1.22) = 1.362 \end{aligned}$$

**TEC** Module 7.3B shows how Euler's method works numerically and visually for a variety of differential equations and step sizes.

This means that if  $y(x)$  is the exact solution, then  $y(0.3) \approx 1.362$ . Proceeding with similar calculations, we get the values in the table:

| $n$ | $x_n$ | $y_n$    | $n$ | $x_n$ | $y_n$    |
|-----|-------|----------|-----|-------|----------|
| 1   | 0.1   | 1.100000 | 6   | 0.6   | 1.943122 |
| 2   | 0.2   | 1.220000 | 7   | 0.7   | 2.197434 |
| 3   | 0.3   | 1.362000 | 8   | 0.8   | 2.487178 |
| 4   | 0.4   | 1.528200 | 9   | 0.9   | 2.815895 |
| 5   | 0.5   | 1.721020 | 10  | 1.0   | 3.187485 |

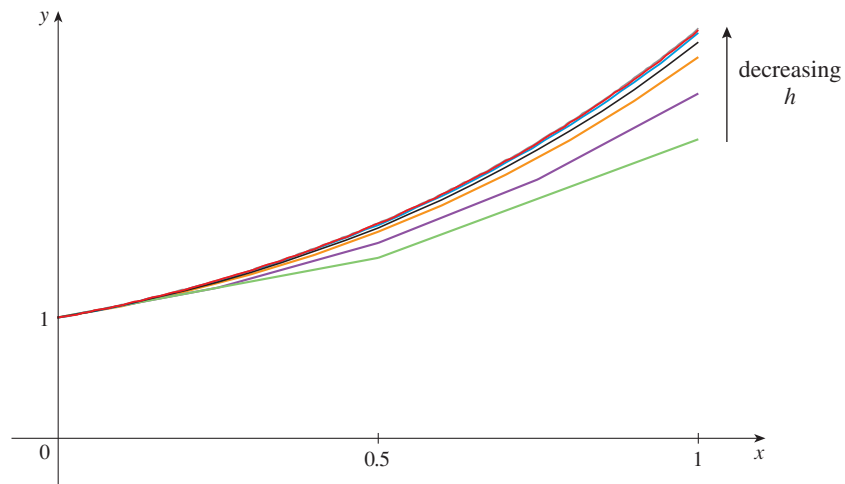
Computer software packages that produce numerical approximations to solutions of differential equations use methods that are refinements of Euler's method. Although Euler's method is simple and not as accurate, it is the basic idea on which the more accurate methods are based.

For a more accurate table of values in Example 3 we could decrease the step size. But for a large number of small steps the amount of computation is considerable and so we need to program a calculator or computer to carry out these calculations. The following table shows the results of applying Euler's method with decreasing step size to the initial-value problem of Example 3.

| Step size | Euler estimate of $y(0.5)$ | Euler estimate of $y(1)$ |
|-----------|----------------------------|--------------------------|
| 0.500     | 1.500000                   | 2.500000                 |
| 0.250     | 1.625000                   | 2.882813                 |
| 0.100     | 1.721020                   | 3.187485                 |
| 0.050     | 1.757789                   | 3.306595                 |
| 0.020     | 1.781212                   | 3.383176                 |
| 0.010     | 1.789264                   | 3.409628                 |
| 0.005     | 1.793337                   | 3.423034                 |
| 0.001     | 1.796619                   | 3.433848                 |

Notice that the Euler estimates in the table seem to be approaching limits, namely, the true values of  $y(0.5)$  and  $y(1)$ . Figure 15 shows graphs of the Euler approximations

with step sizes 0.5, 0.25, 0.1, 0.05, 0.02, 0.01, and 0.005. They are approaching the exact solution curve as the step size  $h$  approaches 0.



**FIGURE 15**  
Euler approximations approaching  
the exact solution

**EXAMPLE 4 | Administering drugs** Estimate the drug concentration (in mg/mL) in the bloodstream after  $\frac{1}{2}$  hour, assuming the concentration changes according to the differential equation

$$\frac{dy}{dt} = 1 + \sin t - y \quad y(0) = 2.5$$

where  $t$  is measured in hours.

**SOLUTION** We use Euler's method with  $F(t, y) = 1 + \sin t - y$ ,  $t_0 = 0$ ,  $y_0 = 2.5$ , and a step size  $h = 0.1$  (which corresponds to six minutes):

$$y_1 = 2.5 + 0.1(1 + \sin 0 - 2.5) = 2.35$$

$$y_2 = 2.35 + 0.1(1 + \sin 0.1 - 2.35) \approx 2.22$$

$$y_3 = 2.22 + 0.1(1 + \sin 0.2 - 2.22) \approx 2.12$$

$$y_4 = 2.12 + 0.1(1 + \sin 0.3 - 2.12) \approx 2.04$$

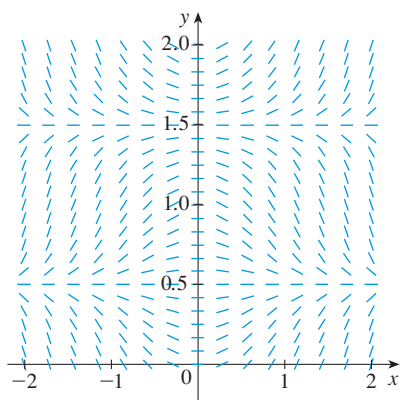
$$y_5 = 2.04 + 0.1(1 + \sin 0.4 - 2.04) \approx 1.97$$

So the concentration after  $\frac{1}{2}$  hour is

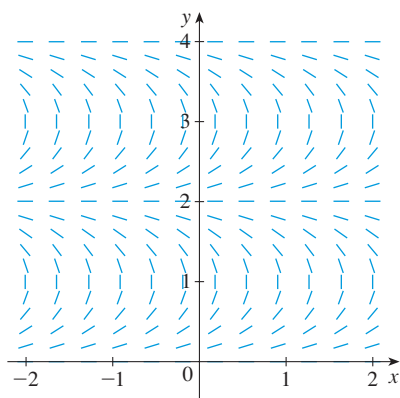
$$y(0.5) \approx 1.97 \text{ mg/mL}$$

## EXERCISES 7.3

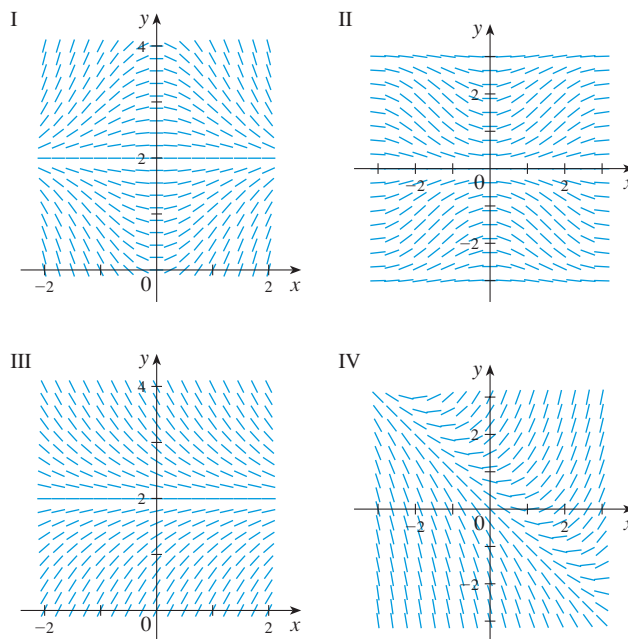
1. A direction field for the differential equation  $y' = x \cos \pi y$  is shown.



- (a) Sketch the graphs of the solutions that satisfy the given initial conditions.
- (i)  $y(0) = 0$                       (ii)  $y(0) = 0.5$
  - (iii)  $y(0) = 1$                       (iv)  $y(0) = 1.6$
- (b) Find all the equilibrium solutions.
2. A direction field for the differential equation  $y' = \tan(\frac{1}{2}\pi y)$  is shown.



- (a) Sketch the graphs of the solutions that satisfy the given initial conditions.
- (i)  $y(0) = 1$                       (ii)  $y(0) = 0.2$
  - (iii)  $y(0) = 2$                       (iv)  $y(1) = 3$
- (b) Find all the equilibrium solutions.
- 3–6** Match the differential equation with its direction field (labeled I–IV). Give reasons for your answer.
- 3.  $y' = 2 - y$                       4.  $y' = x(2 - y)$
  - 5.  $y' = x + y - 1$                       6.  $y' = \sin x \sin y$



7. Use the direction field labeled II (above) to sketch the graphs of the solutions that satisfy the given initial conditions.

(a)  $y(0) = 1$       (b)  $y(0) = 2$       (c)  $y(0) = -1$

8. Use the direction field labeled IV (above) to sketch the graphs of the solutions that satisfy the given initial conditions.

(a)  $y(0) = -1$       (b)  $y(0) = 0$       (c)  $y(0) = 1$

**9–10** Sketch a direction field for the differential equation. Then use it to sketch three solution curves.

9.  $y' = \frac{1}{2}y$

10.  $y' = x - y + 1$

**11–14** Sketch the direction field of the differential equation. Then use it to sketch a solution curve that passes through the given point.

11.  $y' = y - 2x$ ,  $(1, 0)$

12.  $y' = xy - x^2$ ,  $(0, 1)$

13.  $y' = y + xy$ ,  $(0, 1)$

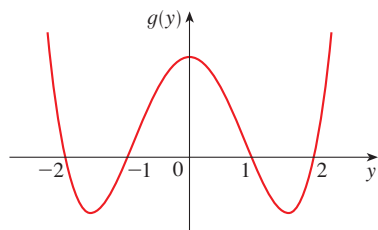
14.  $y' = x + y^2$ ,  $(0, 0)$

**CAS 15–16** Use a computer algebra system to draw a direction field for the given differential equation. Get a printout and sketch on it the solution curve that passes through  $(0, 1)$ . Then use the CAS to draw the solution curve and compare it with your sketch.

15.  $y' = x^2 \sin y$

16.  $y' = x(y^2 - 4)$

- CAS** 17. Use a computer algebra system to draw a direction field for the differential equation  $y' = y^3 - 4y$ . Get a printout and sketch on it solutions that satisfy the initial condition  $y(0) = c$  for various values of  $c$ . For what values of  $c$  does  $\lim_{t \rightarrow \infty} y(t)$  exist? What are the possible values for this limit?
18. Make a rough sketch of a direction field for the autonomous differential equation  $y' = g(y)$ , where the graph of  $g$  is as shown. How does the limiting behavior of solutions depend on the value of  $y(0)$ ?



19. (a) Use Euler's method with each of the following step sizes to estimate the value of  $y(0.4)$ , where  $y$  is the solution of the initial-value problem  $y' = y$ ,  $y(0) = 1$ .  
 (i)  $h = 0.4$       (ii)  $h = 0.2$       (iii)  $h = 0.1$   
 (b) We know that the exact solution of the initial-value problem in part (a) is  $y = e^x$ . Draw, as accurately as you can, the graph of  $y = e^x$ ,  $0 \leq x \leq 0.4$ , together with the Euler approximations using the step sizes in part (a). (Your sketches should resemble Figures 12, 13, and 15.) Use your sketches to decide whether your estimates in part (a) are underestimates or overestimates.  
 (c) The error in Euler's method is the difference between the exact value and the approximate value. Find the errors made in part (a) in using Euler's method to estimate the true value of  $y(0.4)$ , namely,  $e^{0.4}$ . What happens to the error each time the step size is halved?

**20–21 Modeling yeast populations** In Section 7.1 we introduced the following differential equation to describe the dynamics of an experimental yeast population:

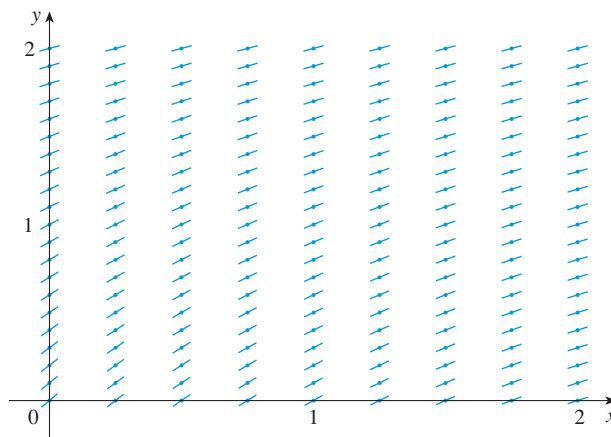
$$\frac{dN}{dt} = (0.55 - 0.0026N)N$$

where  $N(t)$  is the population size (in millions of individuals per mL) at time  $t$  (in hours).

20. Sketch the direction field of the differential equation for values of  $N$  between 0 and 250.
21. Suppose  $N(0) = 0.2$ . Use Euler's method with a step size of  $h = 0.5$  to estimate the population size after four hours. Compare your result to the data in Table 7.1.1.

22. A direction field for a differential equation is shown in the figure. Draw, with a ruler, the graphs of the Euler approximations to the solution curve that passes through the

origin. Use step sizes  $h = 1$  and  $h = 0.5$ . Will the Euler estimates be underestimates or overestimates? Explain.



23. Use Euler's method with step size 0.5 to compute the approximate  $y$ -values  $y_1, y_2, y_3$ , and  $y_4$  of the solution of the initial-value problem  $y' = y - 2x$ ,  $y(1) = 0$ .
24. Use Euler's method with step size 0.2 to estimate  $y(1)$ , where  $y(x)$  is the solution of the initial-value problem  $y' = xy - x^2$ ,  $y(0) = 1$ .
25. Use Euler's method with step size 0.1 to estimate  $y(0.5)$ , where  $y(x)$  is the solution of the initial-value problem  $y' = y + xy$ ,  $y(0) = 1$ .
26. (a) Use Euler's method with step size 0.2 to estimate  $y(0.4)$ , where  $y(x)$  is the solution of the initial-value problem  $y' = x + y^2$ ,  $y(0) = 0$ .  
 (b) Repeat part (a) with step size 0.1.
-  27. (a) Program a calculator or computer to use Euler's method to compute  $y(1)$ , where  $y(x)$  is the solution of the initial-value problem

$$\frac{dy}{dx} + 3x^2y = 6x^2 \quad y(0) = 3$$

for each of the given step sizes.

- (i)  $h = 1$       (ii)  $h = 0.1$   
 (iii)  $h = 0.01$       (iv)  $h = 0.001$
- (b) Verify that  $y = 2 + e^{-x^3}$  is the exact solution of the differential equation.
- (c) Find the errors in using Euler's method to compute  $y(1)$  with the step sizes in part (a). What happens to the error when the step size is divided by 10?
- CAS** 28. (a) Program your computer algebra system, using Euler's method with step size 0.01, to calculate  $y(2)$ , where  $y$  is the solution of the initial-value problem

$$y' = x^3 - y^3 \quad y(0) = 1$$

- (b) Check your work by using the CAS to draw the solution curve.

## 7.4 Separable Equations

We have looked at first-order differential equations from a geometric point of view (phase plots and direction fields) and from a numerical point of view (Euler's method). What about the symbolic point of view? It would be helpful to have an explicit formula for a solution of a differential equation. Although this is not always possible, in this section we examine a commonly encountered type of differential equation that *can* be solved explicitly.

A **separable equation** is a first-order differential equation in which the expression for  $dy/dt$  can be factored as a function of  $t$  times a function of  $y$ . In other words, it can be written in the form

$$\frac{dy}{dt} = f(t) g(y)$$

The name separable comes from the fact that the expression on the right side can be separated into a function of  $t$  and a function of  $y$ . Equivalently, if  $g(y) \neq 0$ , we could write

$$(1) \quad \frac{dy}{dt} = \frac{f(t)}{h(y)}$$

where  $h(y) = 1/g(y)$ . To solve this equation we rewrite it in the differential form

$$h(y) dy = f(t) dt$$

so that all  $y$ 's are on one side of the equation and all  $t$ 's are on the other side. Then we integrate both sides of the equation:

$$(2) \quad \int h(y) dy = \int f(t) dt$$

Equation 2 defines  $y$  implicitly as a function of  $t$ . In some cases we may be able to solve for  $y$  in terms of  $t$ .

We can verify that Equation 2 is indeed a solution using the Chain Rule: If  $h$  and  $f$  satisfy (2), then

$$\frac{d}{dt} \left( \int h(y) dy \right) = \frac{d}{dt} \left( \int f(t) dt \right)$$

$$\text{so} \quad \frac{d}{dy} \left( \int h(y) dy \right) \frac{dy}{dt} = f(t)$$

$$\text{and} \quad h(y) \frac{dy}{dt} = f(t)$$

Thus Equation 1 is satisfied.

One of the simplest applications of the technique of separation of variables is to the differential equation for exponential growth introduced in Sections 3.6 and 7.1. In particular, if  $y(t)$  is the value of some quantity at time  $t$  and if the rate of change of  $y$  with respect to  $t$  is proportional to its size  $y(t)$ , then

$$\frac{dy}{dt} = ky$$

The technique for solving separable differential equations was first used by James Bernoulli (in 1690) in solving a problem about pendulums and by Leibniz (in a letter to Huygens in 1691). John Bernoulli explained the general method in a paper published in 1694.